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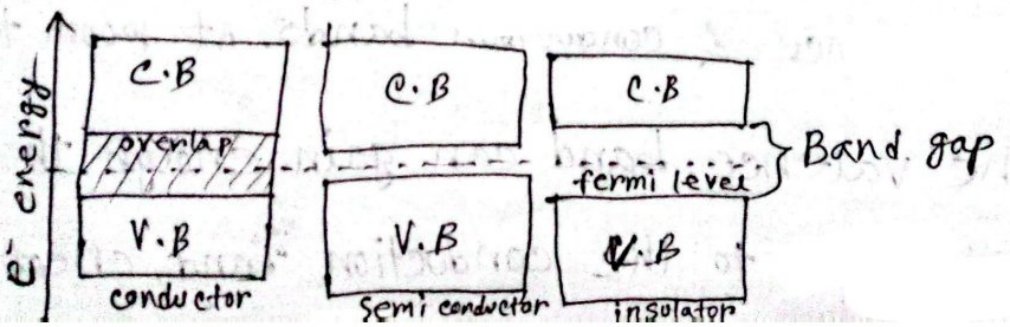
(*) 2016, TU, TU, Based on band theory explain:- (i) fermi level 2021, 2020, 2018, TU

(ii) Forbidden zone 2021 (iii) insulator. (iv) semiconductor

(v) conductor. ?

⇒ (i) Fermi level is an energy level, which lies in the energy band diagram. The highest energy level that an electron can occupy at the absolute zero temperature is known as the fermi level. The fermi level lies btwn the valence band & conduction band because at absolute zero temperature, the e⁻ are all in the lowest energy state. Due to the lack of sufficient energy at 0 Kelvin, the fermi level can be considered as the sea of e⁻ above which no e⁻ exist.



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(II) Forbidden zone: is also known as band gap. It is the energy gap btwn valence band & conduction band in solids: This gap is large in insulators & small in semiconductors. There is no forbidden zone in metals..

(III) Insulator: are materials with large energy gaps btwn the valence & conduction bands. This means that e^- cannot easily move from the valence band (filled with e^-) to the conduction band (where e^- can move freely) resulting in poor electrical conductivity. The large energy gap prevents the flow of e^- even at room Temp., making insulators for electrical insulation purposes. Examples glass, rubber & most plastic.

(IV) semiconductor: are materials with a moderate energy gap btwn valence & conduction bands. At room temp, some e^- in the valence band can gain enough thermal energy to jump to the conduction band, creating charge

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carries (e) & allowing for some electrical conductivity. semiconductors can be doped to increase their conductivity & are vital for electronics, including transistors & diodes. Ex, Si, Ge.

Conductor: are materials with overlapping valence & conduction bands or very small energy gaps btwn them. This allows e⁻ to move freely from the valence band to the conduction band even at room Temp., facilitating high electrical conductivity. Ex, Cu, Ag etc.

2018, 2015, 2017(t), 2020, 2024(t)

Show the difference among conductor, semiconductor & insulator?

	conductor	Semiconductor	insulator
Band Gap	No gap or very small overlap	Moderate gap (around 1 eV)	large gap (greater than 3eV)
Valence band	partially filled	completely filled	completely filled
conduction band	partially filled or overlaps with valence band.	Mostly empty	Empty
conductivity	high	variable: increase with T. or doping	low

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e^- Mobility	High: e^- move easily btwn bands	Moderate: some e^- can move to the conduction band	low: e^- are tightly bound
Resistance	very small	moderate	very high
example	Cu, Ag	Si, Ge	glass, rubber

2018, 2015, 2017 (t), 2021, 2024(t)

⊗ AS Temp. increases resistance also increases of metal?

⇒ with the increase in T, the random motion of e^- increases. As a result number of collitions of e^- with the positive ions increases in a metal. Hence resistance increases.

when a metal is at low T., it's atoms vibrate with low energy and do not disrupt the flow of e^- significantly. However as the T. rises, the atoms vibrate more vigorously due to increased thermal energy. These vibration causing them to collide more frequently & increases resistance.

2016, 2017 (t), 2020

⊗ what are metals? Mention properties of metal? → 2017.

⇒ Metals are a class of materials that have the properties like hardness, lustrous (চকচকে ভাব), malleability (নমনীয়তা),

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ductility and a good conductor of heat & electricity.

⇒ properties of metal:

(I) Good conductor of electricity & heat.

(II) Melleable & ductile, can be molded.

(III) solid at room T., except for mercury.

(IV) have high density & are heavy.

(V) form cation in chemical reaction.

(VI) Have low ionization energy.

(VII) have good tensile strength.

(X) 2018, 2016, TU, TU

Discuss main feature of Band theory of ~~solids~~ metals ?

⇒ In the band theory of metals, the energy level of e^- in a solid material are grouped into bands. The bands represent ranges of energy levels that e^- can occupy. The two main bands are valence & conduction band. This theory explains the quantum state that an e^- takes inside metal solid. Every molecule comprises of various discrete energy levels. The way e^- behave

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inside a molecule is well explained through this theory.

In atoms, molecule are filled in respective energy orbits following pauli's exclusion principle.

In molecules, two atomic orbitals combine together to form a molecular orbital with two distinct energy levels.

In solids, 10^{23} stacked up lines confined in a tiny space would look like a band. Thereby forming energy continuum called energy bands.

This theory helps to visualise the difference btwn conductor, semiconductor, & an insulator. It also helps to explain their unique properties such as electrical conductivity, thermal conductivity and malleability.

2015, 2018, 2017 (t), 2020

⊛ What is doping? what are p & n-type semiconductor?

→ TU

⇒ Doping is the process of adding some impurity atoms in a pure (intrinsic) semiconductor so as to increase the conductivity of a semiconductor. Doping

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can be done in two ways: n-type dopant (pentavalent) is added to an intrinsic semiconductor, to form an n-type semiconductor. when p-type dopant (trivalent) is added then p-type semiconductor is formed.

⇒ when a pure semiconductor is doped with atoms from group III elements such as B, Al or Ga, is called p-type semiconductor.

when pure semiconductor is doped with atoms from group V elements such as P, As or Sb is called n-type semiconductor.

(*) 2015, 2017 (t)
what is called super lattice?

⇒ A super lattice is a layered structure consisting of alternating thin layers of two or more different materials. This layers can be crystalline or amorphous & they typically have thickness on the order of a few nanometers. The key feature of a superlattice is that the properties of the material can be tailored by controlling the thickness & composition of the individua

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layers. It is widely used in semiconductor devices.

(✱) TU What is solid solution?

⇒ A solid solution is a homogeneous mixture of two different kinds of atoms in solid state & having a single crystal structure. Solid solutions can form in various types of crystalline materials including metals, alloys, ceramics etc.

(✱) 2022 Write physical & chemical properties of metal?

⇒ Physical:

① metals are shiny (lustrous) in nature. ② Good conductor of electricity & heat. ③ high density ④ high melting point ⑤ malleable ⑥ Most are ductile (नमनीय) ⑦ opaque (अपारदर्शी) ⑧ High tensile strength ⑨ High B.P. ⑩ Solid except Hg.

Chemical:

① have low electronegetivities ② Good reducing agent ③ Easily corrodible ④ Metals has tend to loss e^- to form cation ⑤ Form Basic oxides ⑥ Form salt with halogens. ⑦ have 1-3 valence e^- ⑧ Many metal react with acid to Produce H-gas & metal salts.

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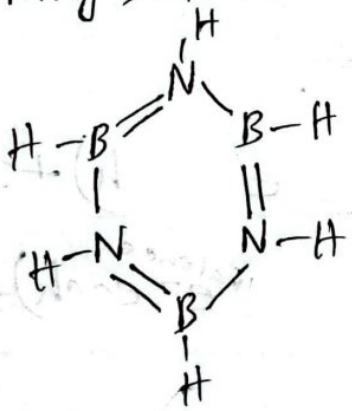
(*) TU. What is borazine? Justify borazine → 2015, 2017 as an electronic isomer of benzene?

⇒ Borazine is a compound also known as "inorganic benzene" or borazole because its structure resemble that of benzene but with B & N atoms in place of carbon atoms. Its chemical formula is $B_3N_3H_6$. Borazine has alternating B & N atoms forming a hexagonal ring, similar to benzene.



⇒ Borazine can be considered as an electronic isomer of benzene because both molecules have the same number of valence e^-.

In benzene, each carbon atom contributes one valence e^-, total six e^- from carbon atoms. these e^- are delocalized throughout the ring structure, forming three alternating double bonds & giving benzene

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its aromaticity.

Similarly, In borazine, each boron atom contribute one valence e⁻, & each N atom contributes five valence e⁻. This totals six valence e⁻ per borazine molecule the same as benzene. These e⁻ in borazine are delocalized around the ring giving it aromatic character.

or Benzen: 6 carbon atoms (4 valence e⁻ each) + 6 H atom (1 e⁻ each) = 30 e⁻

Borazine: 3 boron (3 valence e⁻ each) + 3 N atom (5 valence e⁻ each) + 6 H atom (1 valence e⁻ each) = 30 e⁻.

So, They are iso electric.

2023(t), 2021, 2018, 2017

X What happens when diborane react with

① Excess NH_3 at low T. ⑪ H_2O ⑪ Alcohol ?

⇒ ① $B_2H_6 + NH_3 \xrightarrow[\text{Low Temp.}]{\text{Excess } NH_3} B_2H_6 : 2NH_3$ Ionic

⑪ $B_2H_6 + 3H_2O \longrightarrow B(OH)_3 + 6H_2$

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(iii) $B_2H_6 + 3ROH \longrightarrow B(OR)_3 + 6H_2$

(*) 2023(H) compare Borazine & Benzene ?

Borazine	Benzene
It is inorganic compound having chemical formula $B_3N_3H_6$	It is organic compound having chemical formula C_6H_6.
Inorganic	organic
alternating B & N atom.	only 'C' atom.
comperatively more reactive	Com. less reactive
Not a perfect hexagon	A perfect hexagon
Synthetic	Naturally occurs in crude oil.
aromatic	aromatic

(*) 2015, 2017 What is e- deficient compound ?

⇒ e- deficient compounds are compounds in which an element has an incomplete octet. An e- deficient compound is one that lacks sufficient e- to satisfy the octet rule or achieve a stable electronic configuration.

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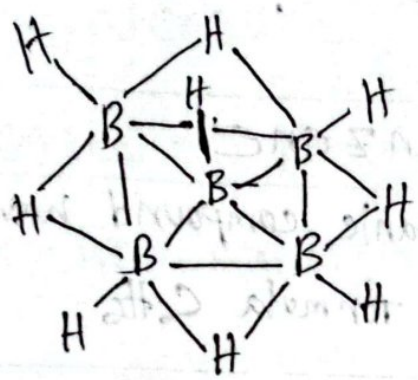
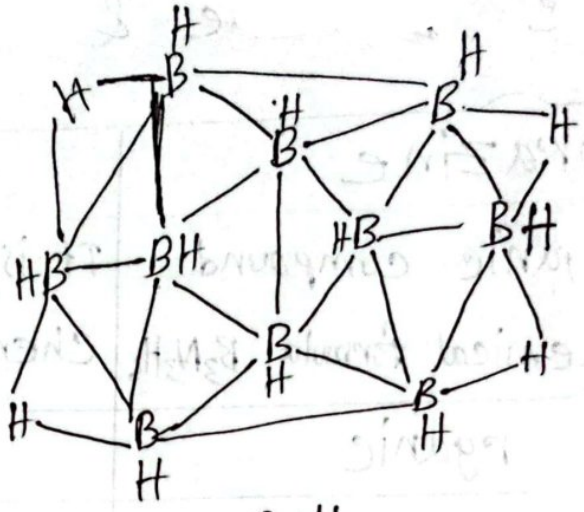
2015, TU, 2023(4), 2022, 2020, 2018, 2029(4)

Draw B_5H_9 & $B_{10}H_{14}$? discuss unusual bonding in diboran



B_5H_9



$B_{10}H_{14}$

⇒ Diborane (B_2H_6), exibits unusual bonding due to its e⁻-deficient nature. Due to its molecular structure & presence of bridging H atom it exibit unusual bonding. The molecule forms "banana bonding" where the boron atoms share e⁻ pairs with each other and with the bridging H atoms. In diborane, each boron atom has an incomplete octat, with only six valence e⁻. TO compensate for this, diborane (ডাইবোরেন) forms 3C-2e bonds, where two boron atoms & a bridging H atom share two e⁻. The bonding

e^- are not localized btwn the two B. atoms but rather spread out, leading to a curved, banana-shaped bond.

2017
Define alloys ?

⇒ Alloys are materials composed of two or more metals or a metal & a non-metal blended together to enhance properties like strength, hardness & resistance to corrosion or heat.

2024(t)
describe intrinsic & extrinsic semiconductor ?

⇒ Intrinsic semiconductors are pure forms of semiconduct

It is a semiconductor without any significant dopant species. Example pure Si, Ge.

⇒ Extrinsic are semiconductors that are doped with specific impurities. The impurity modifies the electrical properties of the semiconductor and makes it more suitable for electronic devices such as diode, transistor.

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We can add ~~dopa~~ two types dopant: pentavalent, trivalent. When we add pentavalent (As, P) dopant then we get n-type semiconductor. When we add trivalent dopant (B, Al) we get p-type semiconductor.

2024(t), 2018, 2020
Why is Borazine called inorganic analogous of benze

⇒ Borazine ($B_3N_3H_6$) is isoelectronic & isostructural with benzene so sometimes is ~~is~~ called inorganic benzene. Borazine is colourless liquid like benzene & gives an aromatic smell. Borazine is an aromatic compound as it follows 4n+2 rule.

This similarity in structure leads to similar properti & reactivity, making it the equivalent of Benzene.

2024(t)
Why B_2H_6 an e⁻ deficient compound?

⇒ B_2H_6 is e⁻ deficient because it lacks enough valence e⁻ to satisfy octate rule. Boron has only three valence e⁻ so diborane cannot satisfy the octate

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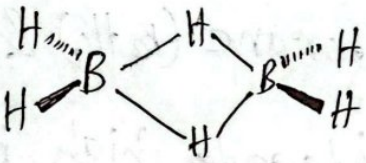
pule for each boron atom in the molecule. B_2H_6 is e⁻ deficient i.e. enough number of e^- are not available to form the expected number of covalent bonds.



or

B_2H_6 is e^- deficient because Boron atom in the molecule only forms two covalent bonds instead of the unusual three bonds that would complete their octate. This deficiency making B_2H_6 more reactive and capable of forming additional bonds with other molecules.

⍟ 2020 How is borazine prepared?

⇒ It can be prepared through several method.

① Reacting B_2H_6 with NH_3 in a 1:2 molar ratio at high Temp. (473K).

$$3\,B_2H_6 + 6\,NH_3 \xrightarrow[\text{ratio } 2NH_3 : 1B_2H_6]{\text{high Temp.}} 2\,\underset{\text{Borazine}}{B_3N_3H_6} + 12\,H_2$$

② $3\,NH_4Cl + 3\,BCl_3 \xrightarrow{140^\circ C} B_3N_3H_3Cl_3 \xrightarrow{[NaBH_4]} \underset{\text{Borazine}}{B_3N_3H_6}$

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(iii) $3H_3BO_3 + 6NH_3 \longrightarrow B_3N_3H_6 + 3H_2O$.

(*) 2021, 2018
Boron hydrides are high potential H-storage materi
Justify the statement ?

⇒ Boron hydride, such as diborane (B_2H_6) & closo-dodecaborate $[B_{12}H_{12}]^{2-}$ are considered high potential H- storage materials due to their high hydrogen content per unit mass. Boron hydrides can store a significant amount of hydrogen by forming stable complexes with H molecules through weak chemical bonding. Also, Boron hydrides have relatively low molecular weights, making them attractive for H-storag applications where weight considerations are critical

(*) 2021
Explain : (i) Electrical conductivity (ii) Malleability (iii) Ductility

⇒ (i) In the e^- gas theory, metals are considered to hav a sea of delocalized e^-. These e^- are not bound to individual atoms but are free to move throughout the metal lattice. when an electric field is applied, the

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mobile e- can flow in response to the voltage, carring electrical current.

Malleability: It refers to a metal's ability to deform under comprehensive stress, typically without breaking. In the e- gas theory, the mobile e- play a crucial role. when a force is applied to a metal, the layers of atoms can slide over each other while the delocalized e- remain relatively free to move. This allows the metal to be reshaped without the bonds btwn atoms breaking, giving it its malleable property.

Ductility: It is the ability of a material to be stretched into a wire. Again, the presence of delocalized e- in the e- gas theory is key. when a metal is pulled ~~str~~ or stretched, the layers of atoms can slide past each other while the mobile e- facilitate this movement. As a result, the metal can be drawn out into a thin wire without

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breaking, demonstrating it's ductile behaviour.

⊛ 2018, 2017, 2015
Show by orbital diagram the formation of 3c-2e open & closed BBB bonds?

⇒ Formation of open BBB bonds (3c-2e⁻): The bond is able to link three B-atoms & is also called Boron bridge bond. Its formation has been depicted as

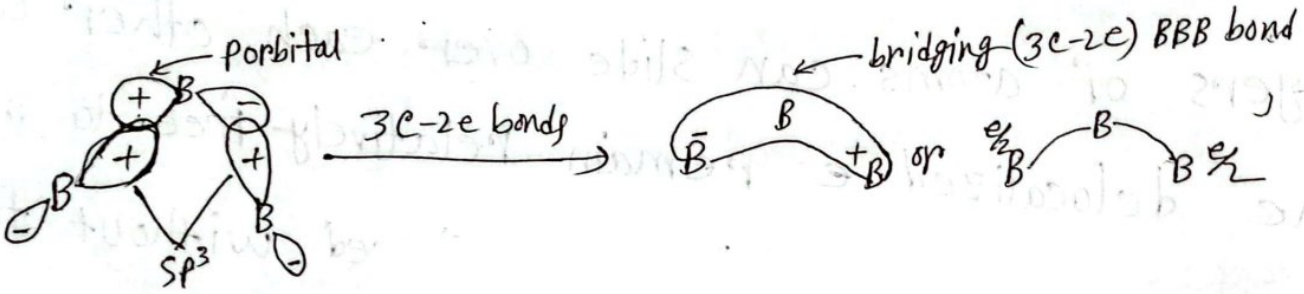

Fig: formation of open bridging (3c-2e) BBB bond.

From figure it is evident that the bridging (3c-2e) BBB bond gets formed by overlap of two hybrid orbitals on two B atoms.

Formation of closed B-B-B bonds (3c-2e⁻): It is formed by the overlap of three hybrid orbitals on B-atoms, which are situated at the 3 centers of an equilateral triangle. Thus this bond each boron atom contributing 2e/3 electrons.

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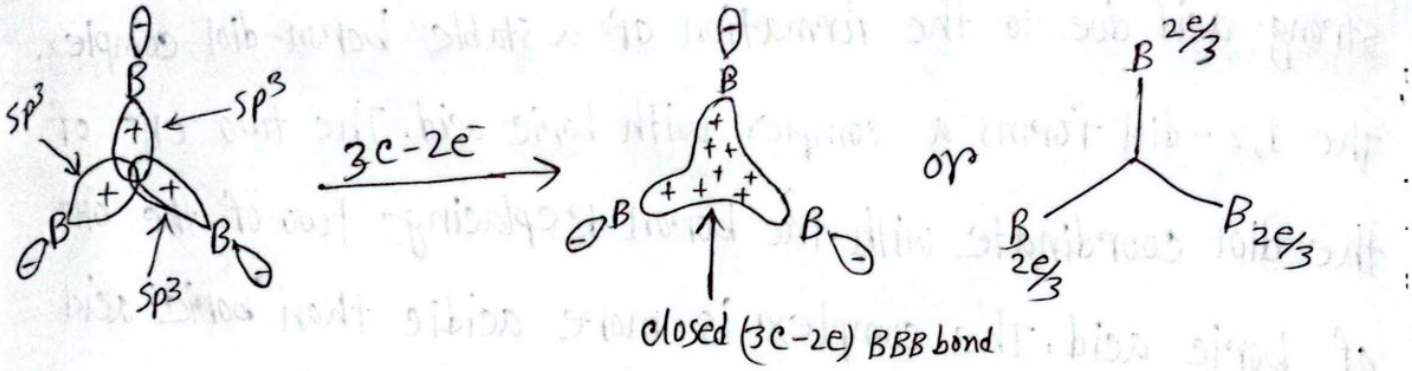

Fig: Formation of closed (3c-2e⁻) BBB bonds

(★) 2015 Explain the order of acidity : $BF_3 < BCl_3 < BBr_3$?

⇒ In boron trihalides, the B has an empty p-orbital. The halogen atoms, can donate, e⁻ density to this empty p-orbital of boron through $p\pi - p\pi$ back bonding. Increasing size of halogen decrease the overlap btwn the p-orbitals of B & the halogen. so the extent of $p\pi - p\pi$ back-bonding decreases. A weaker back-bonding implies less e⁻ density on the B atom, making it more e⁻-deficient & thus stronger lewis acid. So, the order:

$$BF_3 < BCl_3 < BBr_3.$$

(★) 2022, 2021, 2015 Although boric acid is a weak mono basic acid, it behaves as a strong acid in the presence of 1,2-diol - explain?

⇒ H_3BO_3 is a weak monobasic acid in water because it partially ionizes in water, partially react with water & gives H_3O^+ & $[B(OH)_4]^-$. But in presence of 1,2-diols (or; 1,3-diols), boric acid behaves as a

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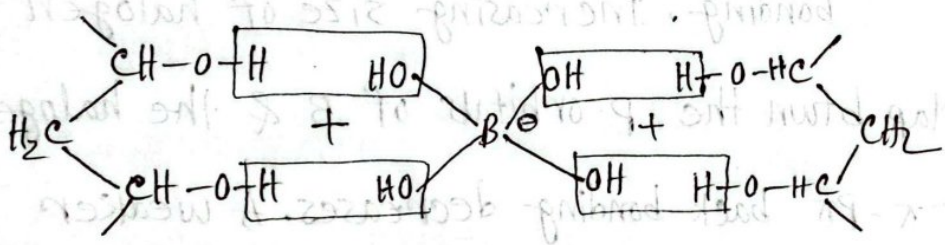
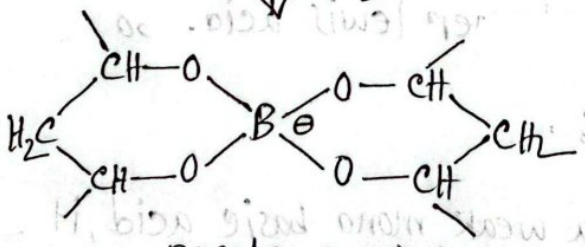
Strong acid due to the formation of a stable boron-diol complex. The 1,2-diol forms a complex with boric acid. The two OH^- of the diol coordinate with the boron replacing two of the OH^- of boric acid. The complex is more acidic than boric acid itself. This is because the cyclic structure of the complex stabilizes the negative charge on the oxygen atom after the proton is released. The increased acidity of the complex leads to a more complete ionization of the proton, making Boric acid behave as a strong acid in the presence of 1,2-diols.

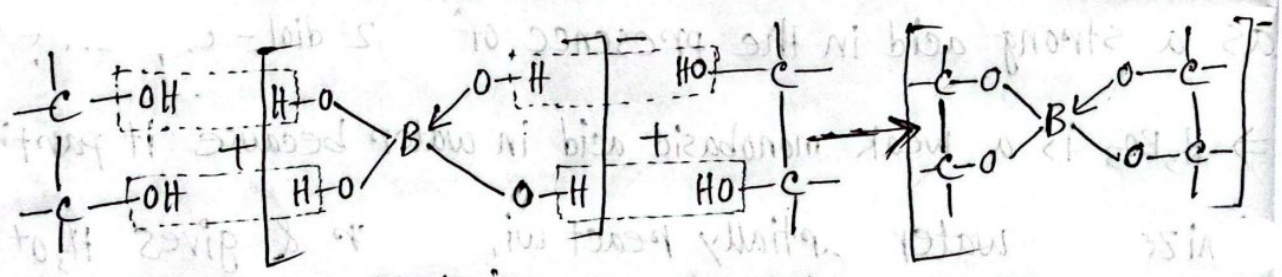


1,3-diol Borate ion 1,3-diol

$-4H_2O$



Borate complex



1,2-diol Borate ion 1,2-diol Borate complex

(X) 2022 when $[Al(BH_4)_3]$ is heated at 70°C, $[Al_2B_4H_8]$ is obtained. write the eqn for this reaction?

⇒ $2[Al(BH_4)_3] \xrightarrow{70°C} [Al_2B_4H_{18}] + B_2H_6$

(X) 2022 what is meant by interstitial & substitutional alloy? Draw the structure for both metal & alloys?

⇒ Interstitial alloys are alloys formed when smaller atoms occupy the interstitial spaces (gaps) btwn the larger atoms in the crystal lattice of a base metal. The smaller atoms are usually much smaller in size compared to the host metal atoms, & their inclusion does not replace the host atoms but fits into the spaces within the metal's structure. (smaller atoms often nonmetals like C, N, H).

Substitutional alloys are alloys in which some atoms of the base metal are replaced (substituted) by atoms of another element. The substituting atoms are typically of smaller size & chemical properties to the atoms of the base metal, allowing them to fit into the crystal lattice without significantly distorting its structure.

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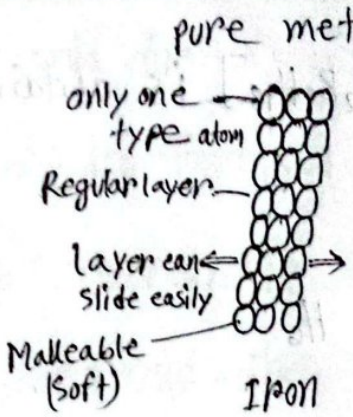
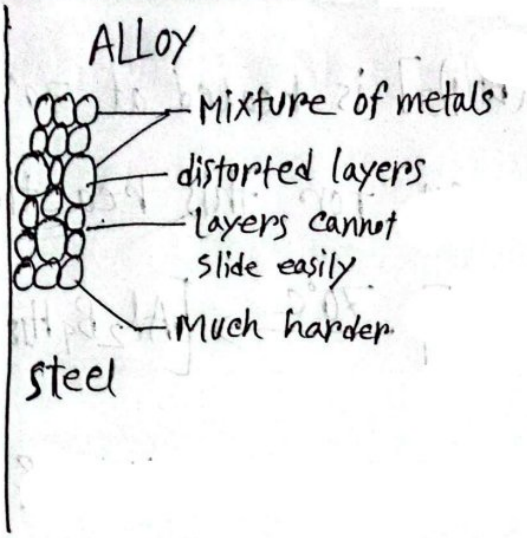
⇒





2022

⊗ Explain why electrical conductivity of a metal decreases as the Temperature raised but opposite occurs with semiconductor?

⇒ metals have a high density of free e^- that are already present at room temperature. These e^- are responsible for conductivity. As temperature increases, the metal atoms vibrate more vigorously. These vibrations disrupt the flow of free e^-, scattering them & hindering their movement. This increased scattering leads to higher resistance & consequently lower electrical conductivity.

semiconductors have a band gap btwn the valence band (where e^- are bound to atoms) & the conduction band (where e^- are free to move). As temperature increases, more e^- gain enough thermal energy to jump from the valence band to

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the conduction band. This increases the number of free charge carries (e & holes), leading to higher conductivity.

(★) 2021, 2016 what are intermetallic compounds? Mention their properties?

⇒ Intermetallic compounds are specific types of chemical compounds formed btwn two or more metals (or a metal & a non-metal), where the elements are present in fixed or nearly fixed ratios. EX. Ni_3Al, CuZn, TiAl etc.

⇒ properties: ① they possess a highly ordered atomic arrangement. ② They have high M.Point. ③ They are generally hard & brittle. ④ some have low density ⑤ They maintain their structural integrity at high temperature. ⑥ The ratio of the elements in the compound is fixed & stoichiometric.

(★) 2021, 2016 what are borohydrides? Draw ① $Be(BH_4)_2$ ② $Al(BH_4)_3$?

⇒ Borohydrides are chemical compounds containing the BH_4^- anion, in which boron is bonded to four H atoms. These compounds are widely used as a reducing agents, particularly for the reduction of carbonyl compounds (Ald & Ketone) to Alcohols.

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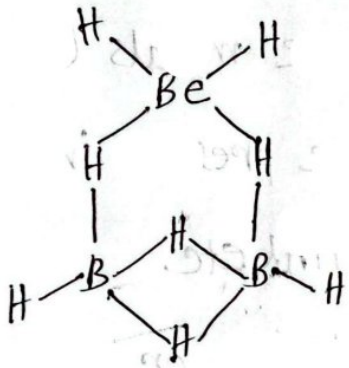
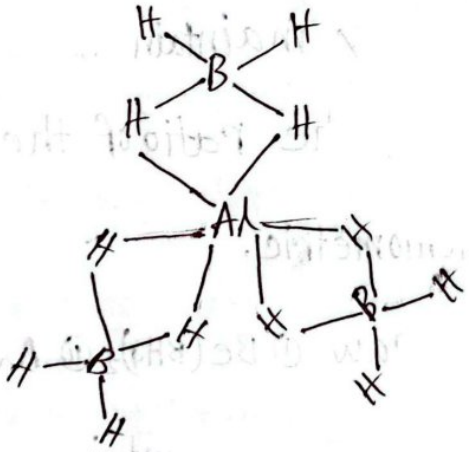
General formula is $M(BH_4)_n$; where M is a metal & n is depends on the oxidation state of the metal. Ex. $NaBH_4$, $LiBH_4$, KBH_4, $Al(BH_4)_3$ etc.

⇒ (I) structure of $Be(BH_4)_2$:



(II) Structure of $Al(BH_4)_3$:



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Non aqueous solvent

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(*) 2021, 2020, TU,
Mention some properties of ionizing solvents?

⇒ (I) Ionizing solvents generally have a high dielectric constant. (II) Many ionizing solvents like water & alcohols, can form H-bond. (III) Ionizing solvents often have a high dipole moment. (IV) Some ionizing solvent can act as acids or bases themselves. (V) They are typically polar solvent (VI) Ionizing power can vary with temperature. (VII) I. solvents can effectively solvate ions, surrounding them with solvent molecules. (VIII) I. solvents can act as e⁻ pair donor & e⁻ pair acceptors.

(*) 2021, → 2017, 2022
what are non aqueous, protic & aprotic solvents?

⇒ non aqueous solvent: is any solvent other than water that is used to dissolved a solute to form a non-aqueous solution. It is not an organic chemical & is not water. ex. liquid NH_3, antimony trichloride, sulfuryl chloride, HF, N_2O_4, pure H_2SO_4, BrF_5 etc.

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Protic solvents: is a solvent that has a H-atom attached to a highly electronegative atom, such as oxygen or Nitrogen allowing it to form hydrogen bonds. Ex. H_2O, HF, NH_3, HCl, H_2SO_4, CH_3OH, C_2H_5OH, CH_3COOH etc.

Aprotic solvents: is a solvent that does not have hydrogen atoms attached to electronegative atoms like oxygen or nitrogen & does not participate in H-bonding. Ex. C_6H_6, CCl_4, CS_2, DMSO, DMF, acetone, CH_3CN, BrF_3, N_2O_4, Tolvene, Hexane, diethyl ether etc. It neither donate nor accept protons.

→ ⊛ 2021, TU, 2015, 2018
Sln of alkali metals in liq. NH_3 is paramagnetic & blue in colour - Explain?

⇒ When alkali metals are dissolved in liq. NH_3 they ionize to give metal ions & valence e^- as:

$$Na \longleftrightarrow Na^+ + e^-$$

Both alkali metals & e becomes solvated by NH_3 molecules.

$$Na^+ + xNH_3 \longrightarrow [Na(NH_3)_x]^+$$

(ammoniated cation)

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$$e^- + yNH_3 \longrightarrow [e(NH_3)_y]^-$$

ammoniated e^-

the complete reaction may be written as:

$$Na \longrightarrow [Na^+(NH_3)_x]^+ + [e(NH_3)_y]^-$$

The ammoniated e^- are responsible for blue colour of sln.

$$M(s) + NH_3(l) \longrightarrow [M(NH_3)^+] + e^-(solvated)$$

solvated e^- are unpaired e^-, making the sln paramagnetic. the blue colour arises from the absorption of light by the solvated e^-. these e^- absorbs light in the red part of the visible spectrum, allowing the blue light to be transmitted, resulting in the characteristic blue color.

$$M \xrightarrow{\text{dissolve in liq. } NH_3} M^+(solvated) + e^-(solvated)$$

2017, 2015, 2022, 2021, 2018

⊛ acetic acid is weak acid in aqueous sln but behaves as a strong acid in liq. NH_3 - explain why?

⇒ In aqueous sln CH_3COOH does not completely dissociate into its ions (H^+ & CH_3COO^-). water can act as both an acid

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and a base & water can produce H_3O^+ ions readily, acetic acid does not fully ionize. ~~so CH_3COOH~~

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

The equilibrium constant (K_a) for this reaction is relatively small, indicating that acetic acid is a weak acid in water.

When acetic acid dissolved in liq. NH_3, CH_3COOH donates proton to NH_3, forming CH_3COO^- & NH_4^+. Because NH_3 has a much lower tendency to ionize, it readily accepts protons from acetic ~~acid~~ acid more completely, making it behaves as a strong acid in this Medium.

$$CH_3COOH(l) + NH_3(l) \rightleftharpoons CH_3COO^-(\text{solvated}) + NH_4^+(\text{solvated})$$

'K_a' for this reaction is much larger than H_2O, indicating that acetic acid acts as a strong acid in NH_3.

⊗ 2018, 2017, ~~2016~~ complete the following reaction ?

⇒ ① $PCl_5 + SO_2$ (Liq) $\longrightarrow POCl_3 + SOCl_2$ → 2016, TU

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(II) 2016 $2FeCl_3 + 2KI \xrightarrow{Liq.\ SO_2} 2FeCl_2 + 2KCl + I_2$

(III) $HClO_4 + H_2SO_4 \xrightarrow{Liq.\ H_2SO_4} H_3SO_4^+ + ClO_4^-$

(IV) $CH_3COOH + H_2SO_4\ (Liq) \longrightarrow CH_3COOH_2^+ + HSO_4^-$

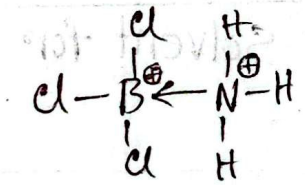
(V) $NH_2CONH_2 + H_2SO_4\ (Liq) \longrightarrow NH_2CONH_3^+ + HSO_4^-$

(X) 2016, TU complete the following reaction ?

⇒ (I) TU $KI + NH_4Cl \xrightarrow{Liq.\ NH_3} NH_4I + KCl$

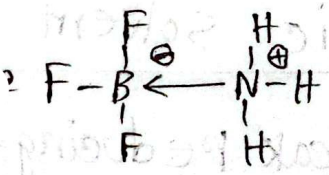
(II) $NH_4Cl + KNH_2 \xrightarrow{Liq.\ NH_3} KCl + 2NH_3$

(III) $BCl_3 + NH_3 \xrightarrow{Liq.\ NH_3}$



(IV) $SbCl_3 + 3KCl \xrightarrow{Liq.\ SO_2}$ $K_3[SbCl_6]$

(V) $BF_3 + NH_3\ (Liq) \longrightarrow$



(VI) $PH_3 + N_2 \xrightarrow{Liq.\ NH_3}$ No reaction

(VII) $2KI + SOCl_2 \xrightarrow{Liq.\ SO_2} 2KCl + SO_2 + I_2$

(VIII) 2021 $BiI_3 + KNH_2 \xrightarrow{Liq\ NH_3} BiN + 3KI + 3NH_3$

(IX) 2021 $6KI + 3SbCl_5 \longrightarrow 2K_3[SbCl_6] + SbCl_3 + 3I_2$

(*) 2020, 2018, 2016, TU
Compare the solvent properties H_2O & liq. NH_3?

⇒

H_2O	Liq. NH_3
H_2O has strong H-bonding due to the O-H bond.	Liq. NH_3 form weaker H-bond than H_2O
H_2O is highly polar solvent than NH_3	NH_3 is lower polar solvent than water
Boiling point of H_2O is high which is 100°C.	B.P. of liq. NH_3 is lower than H_2O which is −33°C
H_2O is weak base	NH_3 is strong base
liquid at room temp.	liquid at low temp. (−33°C).
excellent solvent for ionic compounds	Good solvents for alkali metals.
Dielectric constant is high	D.C. is low
Amphoteric solvent	Basic solvent
Relatively weak reducing agent	strong reducing agent
universal solvent	specialized solvent in organic synthesis

(*) 2016
What are the properties of solvent? How are solvent classified?

⇒ A solvent is the material present in the larger

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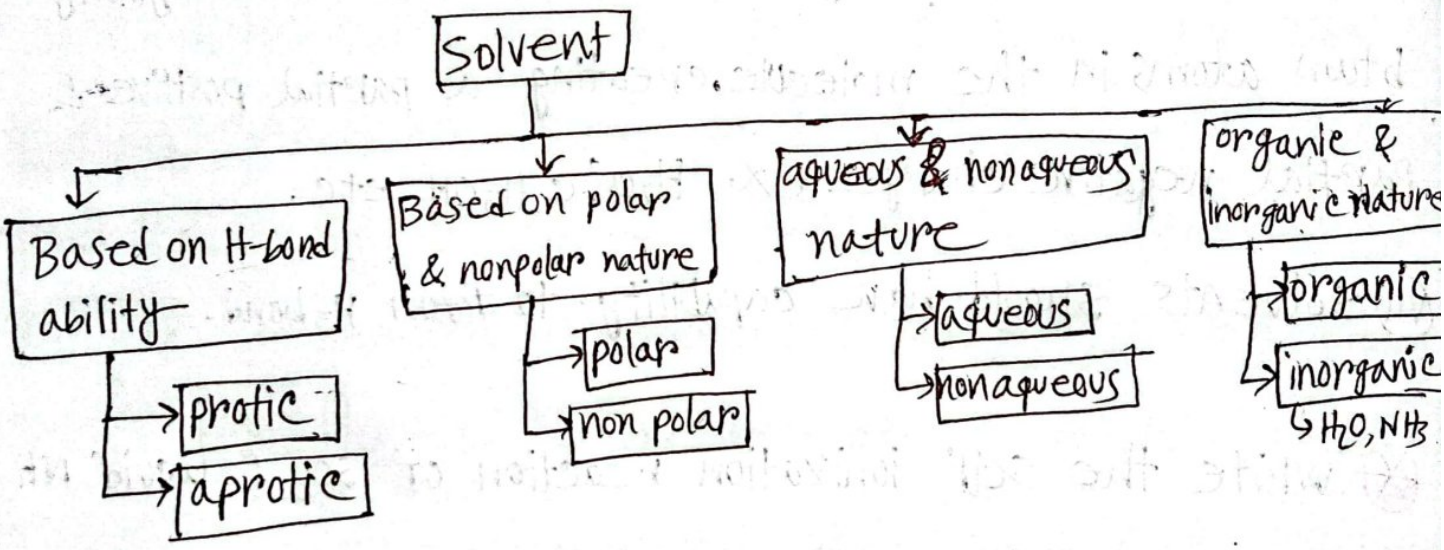
amount in the sln & is responsible for dissolving the solute. properties of solvents are: ① The Temp. range over which it is a liquid. ② Its dielectric constant. ③ Its donor & acceptor properties ④ Its protonic acidity & basicity ⑤ H- bonding ⑥ the nature & extent of auto dissociation.

⇒ classification of solvents:



⑧ 2015, 2018 What are the features of a substance to be a solvent?

⇒ A solvent is the material present in the larger amount in the sln & is responsible for dissolving the solute. Feasures:

① It should have dielectric constant.

② It should have protonic acidity & basicity.

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(III) It should have donor & acceptor properties.

(IV) The solvent should have a wide range of Temp. over which it is a liquid.

(V) It should have the nature & extent of autodissociation.

$$H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$$

$$NH_3 + NH_3 \rightleftharpoons NH_4^+ + NH_2^-$$

(VI) The solvents has a significant difference in electronegtivity btwn atoms in the molecule, creating a partial positive & partial negative charge. Ex. H_2O; C_2H_5OH etc.

(VII) Solvents should have capability to form H-bond.

(X) TU Write the self ionization reaction of SO_2 & liquid NH_3 with example?

⇒ Self-ionization (or autoionization) is a process where a molecule acts as both and a base, reacting with itself to form ions.

$$2SO_2 \rightleftharpoons SO^{2+} + SO_3^{2-}$$

two SO_2 molecules react to form a solvated sulfuryl cation (SO_2^{2+}) and a solvated sulfite anion (SO_3^{2-}).

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In liquid NH_3, a molecule of NH_3 can also act as both an acid ~~&ref~~ and a base.

$$2NH_3 \rightleftharpoons NH_4^+ + NH_2^-$$

one NH_3 molecule act as a proton donor & forms NH_4^+. other NH_3 act as a H^+ acceptor & forms NH_2^- (amide ion).

2021

(*) Write a brief note on ionic liquid as solvent?

⇒ Ionic liquids are salts that are liquid at or near room temperature (often below 100°c). Ionic liquids are composed entirely of ions. Many I.L. exibit excellent thermal stability. IL can act as solvents, catalysts or both, enhancing r. rate. IL are used as electrolytes in batteries, fuel cells. IL are used in the synthesis of nanomaterials, polymers. IL have negligible vapour pressure, making them non-volatile. This also reduces environmental pollution. IL can dissolve a wide range of compounds, including polymers, inorganic salts, & biomolecules. Due to their low volatility & potential for recyclability, IL are considered more environmental freindly. Generally

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Ionic liquids(IL) are Green solvents, non-flammable, very low vapour pressure, High thermal stability, High ion conductivity.

2020

The dissolution of alkali metal in liq. NH_3 produce a: (I) Blue coloration (II) is a good conductor of electricity?

⇒

(I) The blue colouration that appears when alkalimetals dissolve in $NH_3(liq)$ is due to the formation of solvated e^-

$$M(s) + NH_3(l) \longrightarrow [M(NH_3)^+] + e^-(solvated)$$

The blue colour arises from the absorption of light by the solvated e^-. These e^- absorbs light in the red part of the visible spectrum, allowing the blue light to be transmitted, resulting in the characteristic blue colour.

$$M \xrightarrow{\text{dissolve in liq. } NH_3} M^+(solvated) + e^-(solvated)$$

(II) The dissolution of alkali metals in liq. NH_3 produces

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a good conductor of electricity due to the presence of solvated e^-

$$M \xrightarrow{\text{dissolve in liq. } NH_3} M^+(\text{solvated}) + e^-(\text{solvated})$$

This solvated e^- are highly mobile & can move freely throughout the sln. This mobility allows the sln to conduct electricity.

2018, 2017

Urea shows no acidic behaviour at all in H_2O but behaves as a ~~strong~~ weak acid in NH_3 ?

⇒ water can act as both acid and base. water is a relatively weak base, so it doesn't accept protons ~~fom~~ from urea. so urea doesn't exibit acidic behavior in H_2O. also H_2O is much stronger proton donor than urea, so urea remains in its neutral form.

NH_3 is stronger base than H_2O. It readily accepts protons. urea can act as a weak acid in NH_3 by donating H^+ to the NH_3 molecule.

$$NH_2CONH_2 + NH_3(liq) \longrightarrow NH_2CONH^- + NH_4^+$$

Urea donates H^+ to NH_3, forming NH_4^+ (ammonium ions), that's

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Why urea ($NH_2-CO-NH_2$) behaves as a weak acid in NH_3

2022
Anhydrous H_2SO_4 is highly viscous & has low ion mobilities - explain?

⇒ Because of its strong intermolecular H-bonding & lack of dissociation in the absence of H_2O. Each H_2SO_4 can form multiple H-bonds with neighboring molecules, creating a highly associated network. This extensive H-bonded network makes the liquid structure more rigid, resulting in high viscosity. In absence of H_2O, H_2SO_4 remains as neutral molecules rather than forming H_3O^+ & HSO_4^- ions. So without dissociation there are very few ions to carry a charge, leading to low ion mobility.

The HSO_4^- & $H_3SO_4^+$ ions formed during self ionization are relatively large & bulky, which also contributes to lower mobility.

2018, 2017
Solvolysis of Cl_2 in NH_3 is parallel to that in H_2O?

⇒ because both reactions involve the solvent acting as a nucleophile to react with Cl_2, resulting in analogous समान

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products. Both H_2O & NH_3 act as a nucleophiles, donating a lone pair to facilitate the reaction. Both involve disproportion reaction where chlorine undergoes both oxidation and reduction.

Solvolysis in water : $Cl_2 + H_2O \longrightarrow HCl + HOCl$

Solvolysis in ammonia: $Cl_2 + 2NH_3 \longrightarrow NH_4Cl + NH_2Cl$

Chlorine undergoes similar cleavage in both solvents in a similar fashion. Both Reaction produce an acid & a base. This parallelism arises because both solvents have lone pairs of e^- capable of nucleophilic attack & breaking Cl–Cl bond in similar fashion.

2016

Addition of alkaline earth metal in liq. NH_3 produces a complex, whereas the sln of alkali metal in liq. NH_3 is coloured - explain ?

⇒ When alkali metal dissolve in liq. NH_3, they undergo ionization to form metal cation (M^+) & solvated e^-, which are surrounded by NH_3 molecules. This solvate

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e^- are responsible for the deep blue colour of sln due to their ability to absorb light & undergo electronic transitions. The color intensity depends on the conc. of solvated e^-. At higher conc. the sln becomes bronze-coloured.

$$M \xrightarrow{\text{dissolved in liq. } NH_3} M^+(\text{solvated}) + e^-(\text{solvated})$$

Alkaline earth metal have two valence e^-, having higher ionization energy & being divalent (M^{2+}), tent to form more stable complexes in liq. NH_3 rather than producing free solvated e^-. These metals usually react with NH_3 to form complexes, where metal cation is coordinated with NH_3. Due to the lack of free solvated e^- the sln is not colored. A. earth metal, with their higher charge density, interact more strongly with the lone pair of e^- on the N-atom of NH_3. This leads to the formation of metal-ammoni complexes.

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2015

⊗ Seperate sln of liq. SO_2 in H_2O ~~but~~ & SO_2 in HBr cannot conduct electricity but mixture of two sln is a good conductor of electricity - explain?

⇒ H_2SO_4 is only sparingly soluble in H_2O & form H_2SO_3 which does not ionize significantly.

$$SO_2 + H_2O(l) \rightleftharpoons H_2SO_3(aq) \rightleftharpoons H^+(aq) + HSO_3^-(aq)$$

This results a low concentration of H^+ ions & HSO_3^- ions. Due to low concentration of ions, the sln is a poor conductor of electricity.

SO_2 wouldn't react significantly with HBr. Neither SO_2 nor HBr ionizes significantly in this sln. As there is no free ions to carry the electrical charge, the sln cannot conduct electricity.

When two sln are mixed, ~~a redox reaction occurs btwn H_2SO_3 and HBr~~. SO_2 reacts ~~for~~ with HBr & water, producing H^+ & Br^-.

$$SO_2 + 2HBr + H_2O \longrightarrow H_3O^+ + Br^- + HSO_3^-$$

Because of mobile Br^-, H_3O^+ ion it conduct electricity.

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(X) 2022

Show the following reactions both in liq. NH_3 & H_2O:

(I) Self-ionization (II) Amphoteric behavior of $[Zn(OH)_2]$?

⇒ In water, $Zn(OH)_2$ can undergo self ionization ~~as follows:~~ in two ways, demonstrating its amphoteric behaviour:

(1) As an acid: $Zn(OH)_2 + 2H_2O(l) \rightleftharpoons [Zn(OH)_4]^{2-} + 2H^+$

(2) As a base: $Zn(OH)_2(s) \rightleftharpoons Zn^{2+} + 2OH^-$

In liquid NH_3, $Zn(OH)_2$ can undergo self ionization as follows:

(1) As an acid: $Zn(OH)_2 + 4NH_3(l) \rightleftharpoons [Zn(NH_3)_4]^{2+} + 2OH^-$

⇒ (II) $Zn(OH)_2$ demonstrates amphoteric behavior in water by reacting with both acid & bases:

When $Zn(OH)_2$ react with acid, it accepts proton (H^+) & OH^- ion from $Zn(OH)_2$ combine with H^+ from the acid to form water.

$$Zn(OH)_2 + 2H^+ \longrightarrow Zn^{2+} + 2H_2O$$

EX. $Zn(OH)_2 + 2HCl \longrightarrow ZnCl_2 + 2H_2O$

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When $Zn(OH)_2$ reacts with a base it forms complex ion called zincate, $[Zn(OH)_4]^{2-}$.

$$Zn(OH)_2(s) + 2OH^-(aq) \rightleftharpoons [Zn(OH)_4]^{2-}(aq)$$

Ex. $Zn(OH)_2 + 2NaOH(aq) \rightleftharpoons Na_2[Zn(OH)_4](aq)$

In liq. NH_3, $Zn(OH)_2$ exhibits amphoteric behavior, reacting with acidic and basic species.

① As an acid: $Zn(OH)_2 + 4NH_3(l) \longrightarrow [Zn(NH_3)_4]^{2+} + 2OH^-$

② As a Base: $Zn(OH)_2 + 2NH_4Cl \longrightarrow ZnCl_2 + 2NH_3 + 2H_2O$

2017.

Addition of Alkali metal to NH_3 creat blue sln with electrical conductivity comparable to metal. Explain the following: ① Solvated e^- theory

② blue sln is paramagnetic

③ blue sln is a good one-e^- reducing agent

⇒ ① Solvated e^- theory: It explains the behavior of e^- that are free but surrounded by solvent molecules, and that is stabilized by solvation. When alkali metal

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like K/Na are dissolved in liquid NH_3, they lose an e^- to form cation. The released e^- is solvated by NH_3, forming a blue soln. solvated e^- theory is used in organic synthesis, Radiation chemistry, Electrochemistry.

$$M \xrightarrow{\text{dissolved in Liq. } NH_3} M^+_{(solvated)} + e^-_{(solvated)}$$

the released e^- is stabilized by the surrounding NH_3 molecules. this e^- is called solvated e^-.
Solvated e^- is responsible for color, conductivity, Reducing power, paramagnetism etc.

⇒ (ii) Solvated e^- absorbs light in the red part of the visible spectrum, allowing blue light to be transmitted, resulting in the blue color soln we get. Alkali metal has one valence electron & they produce unpair (one) solvated e^-. The paramagnetism arises from presence of unpaired e^-. These unpaired e^- possess a magnetic moment and are attracted to an external magnetic field. The more unpaired e^-, the stronger paramagnetism. Since solvated e^- in blue soln is unpaired so the soln

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exhibits paramagnetism.

(iii) Due to the presence of solvated e^- which are highly reactive & possess unique properties that make them excellent at donating e^- in redox reaction. The solvated e^- is typically donate only one e^- at a time. This makes it a one-e^- reducing agent. Solvated e^- is highly reactive species. It can easily donate its e^- to another substance, reducing it. This makes the s/n a strong reducing agent. The e^- is free to reduce other species by transferring a single e^-, so it is one-electron reducing agent.

2023(t)

(*) what is auto-ionization?

⇒ It is a process in which molecules of a substance react with each other, producing both a cation & an anion, without the involvement of any external reagents.

Ex. $H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$

In this reaction, one water molecule donates a H^+ to another water molecule, forming H_3O^+ & OH^-.

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other examples:

① $2NH_3 \rightleftharpoons NH_4^+ + NH_2^-$

② $2H_2SO_4 \rightleftharpoons H_3SO_4^+ + HSO_4^-$

TU(2017)

⊛ Define solvolysis & solvation reaction with example?

⇒ Solvolysis is a chemical reaction where the solvent act as a nucleophile & attacks a substrate molecule leading to a substitution or elimination reaction. If the solvent is water, the process is called hydrolysis; if it is alcohol, it is called alcoholysis & so on. General reaction:

$$RX + \text{Solvent} \longrightarrow R\text{-Solvent} + X^-$$

EX. $(CH_3)_3CBr + H_2O \longrightarrow (CH_3)_3COH + HBr$

⇒ A solvation reaction is a chemical process in which solute particles interact with solvent molecules, leading to the stabilization of the solute species in the sln. EX: Dissolving NaCl in H_2O, Dissolving O_2 in hexane.

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2017(t)

⊗ In liquid NH_3 soln the strong oxidizing agent aren't stable. Why?

⇒ Because the solvent, liq. NH_3 is a relatively strong reducing agent. NH_3 contain lone pair of e^- that can be donated, readily reducing oxidizing agents. The oxidizing agent when dissolved in liq. NH_3, can be reduced by the NH_3. NH_3 readily provide e^- to oxidizing agents, reducing them & destabilizing their oxidizing forms.

2023(t)

⊗ Why Non-aqueous solvents are preferred over H_2O?

⇒ (I) Some solutes, particularly organic or non-polar compounds, do not dissolve well in water due to its high polarity & H-bonding nature. Non aqueous solvents can dissolve these substance.

(II) Non-aqueous solvents often have lower freezing points & higher boiling points than water, allowing reactions to be carried out under extreme temperature condition.

(III) Water is reactive & can participate in hydrolysis, oxidation, or reduction reaction. Non aqueous may be chemically inert, making them more suitable for reactions where water

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would interfere.

(IV) water is amphoteric, this can lead to unwanted side reactions. Non-aqueous solvents with specific acidic, basic or neutral properties can be used to control the reaction environment.

(V) Reactions in non aqueous solvents can have different Temp. dependencies, which can be advantageous for specific reactions.

(VI) Non aqueous can provide unique reaction environments that promote specific chemical reactions.

(VII) Non-aqueous exhibit unique solvating properties enabling reactions.

2023(t)

(X) Define: (i) amphi-protic solvent (ii) ionizing solvent ?

⇒ amphi-protic: is a solvent that can act both as a proton donor (acid) & as a proton acceptor (base). It can act as protophilic or protogenic in nature. EX. H_2O, NH_3, Alcohol etc

(ii) ionizing solvent: is a solvent that has the ability to dissociate solute molecules or ions into their constituent

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ions. This dissociation process is known as ionization.

EX. H_2O, liq. NH_3, Alcohol, acetic acid etc.

2023(t)

⊗ Differentiate btwn Solvation & solvolysis reaction with example?

Solvation	Solvolysis
The process of solute molecules being surrounded by solvent molecules.	A C. reaction where solvent act as a nucleophile, attacking a ~~nucleophile~~ substrate & causing a bond to break.
NO bond breaking occur	A bond in the substrate molecule breaks.
Solvent molecules surrounded the solute without reacting with it.	Solvent molecules directly participate in the reaction as a nucleophile
Physical interaction leading to Stabilization of solute particles.	chemical r. resulting in the cleavage of bonds in the solute.
Act as a stabilizing medium for ions or molecules.	Act as a reactant
Ex. NaCl dissolving in H_2O	Ex. $(CH_3)_3CBr + H_2O \rightarrow (CH_3)_3COH + HBr$
Generally reversible	Generally irreversible under normal condition

2023(t)

⊗ Show auto-ionization for the following solvents & identify conjugate acid & base : NH_3, SO_2 and H_2S ?

⇒ for NH_3 : $2NH_3(l) \rightleftharpoons \underset{\text{conjugate acid}}{NH_4^+} + \underset{\text{conjugate base}}{NH_2^-}$

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For SO_2: $2SO_2 \rightleftharpoons SO^{2+} + SO_3^{2-}$

c. acid c. base

For H_2S: $2H_2S \rightleftharpoons H_3S^+ + HS^-$

c. acid c. base

TU (2023)

Q. What are "Ammoniated Metal" & "ammoniated e^-"?

⇒ Ammoniated Metal: is a metal cation that dissolves in liquid NH_3 & that is solvated by NH_3 molecules. The NH_3 molecules act as ligands, forming coordinate covalent bonds with the metal ion.

Ammoniated e^-: is a solvated e^- formed when metals dissolve in liquid NH_3. It is free e^- that is surrounded by NH_3 molecules.

$$M \xrightarrow{\text{dissolve in liq. } NH_3} M^+_{(\text{ammoniated})} + e^-_{(\text{ammoniated})}$$

2023 (t)

Q. Alkali metal in liq. NH_3 are all blue in colour, the absorption spectra are alike & the blue sln are excellent electrolyte-explain?

⇒ solvated e^- absorbs light in the red part of the visible spectrum. This absorption leaves the complemental

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color blue, which is transmitted, giving the sln its characteristic blue color. different alkalis metal also produce one solvated e⁻ so all sln are blue color. The absorption spectra are same (alike) for all alkali metal because the absorption process primarily involves the solvated e⁻ which are identical for all alkali metal.

Solvated e⁻ are highly mobile, allowing them to conduct electricity. This high mobility, coupled with the presence of both cation & anions in the sln, makes these sln excellent electrolytes.

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Inorganic polymers

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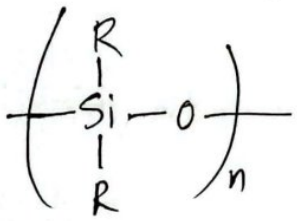
(★) 2018, 2017, TU → 2022 What is Silicone polymer? Discuss the preparation of Silicones?

⇒ Silicone polymers also known as polysiloxanes or silicones, are a group of synthetic polymers made primarily of silicon, oxygen carbon & hydrogen. These polymers have a unique backbone structure consisting of repeating siloxane units $(-O-SiR_2-O-SiR_2-$ where R is alkyl chain. These polymers are unique due to their inorganic silicon-oxygen backbone. This can be represented by general formula $(R_2SiO)_n$, whose structure,



The structures shows that silicones have Si-O & Si-C bonds.

⇒ preparation: The starting material for the manufacture of silicons are alkyl or aryl substituted chlorosilen.

$$R-Cl + Si \xrightarrow[300^\circ C]{Cu} R_3SiCl + R_2SiCl_2 + RSiCl_3$$

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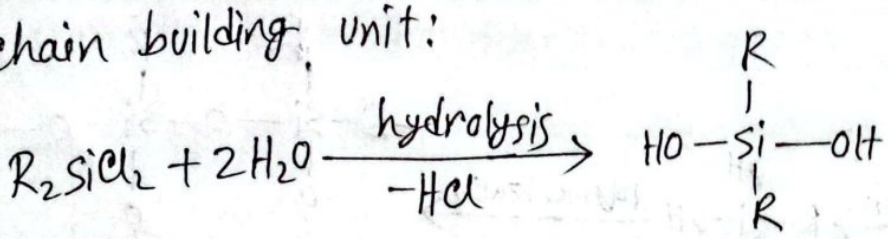
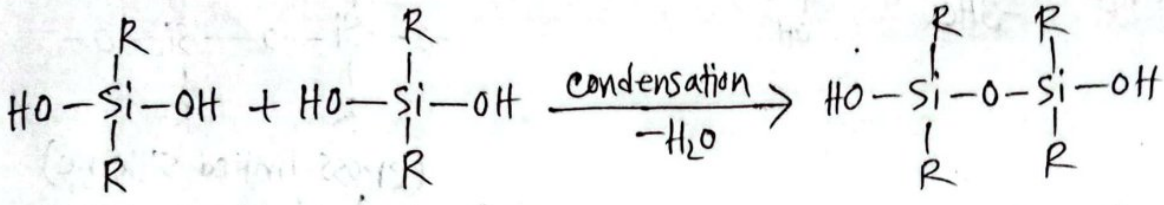
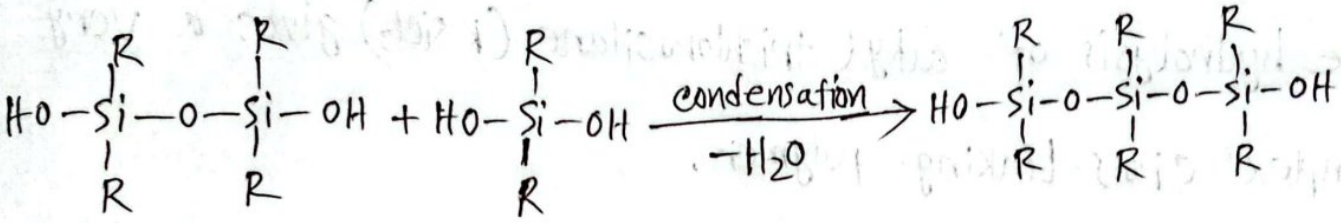
① Chain building unit:

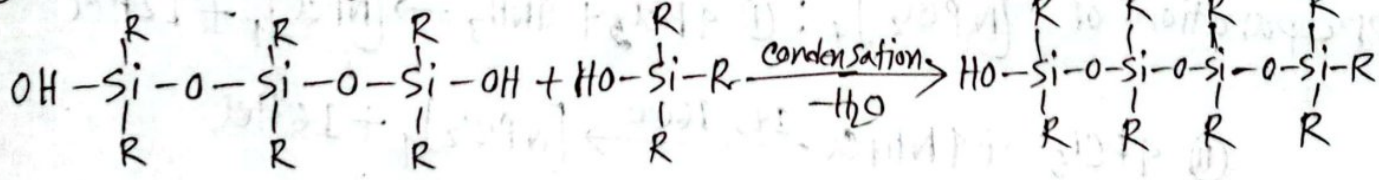






Here, R_2SiCl_2 gives rise to straight chain polymers & as having an active -OH group at each end of chain, polymerization continues & chain increases in length. So, R_2SiCl_2 is therefore a chain building unit.

② Chain stopping unit.



If some R_3SiCl is mixed with R_2SiCl_2 & hydrolised, then R_3SiCl will block one end of the straight chain produced by R_2SiCl_2. Since there is no active OH group at the end of the chain, the chain stops here. Eventually the other end can be blocked in similar way. Thus R_3SiCl is a chain stopping

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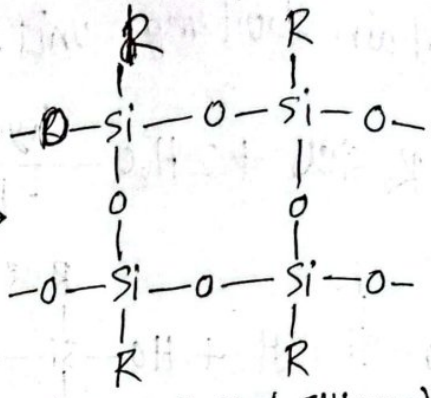
unit.

(iii) cross linking unit:

$$RSiCl_3 + 3H_2O \xrightarrow[-3HCl]{\text{hydrolysis}} R-Si(OH)_3 \xrightarrow{\text{polymerization}}$$



(cross linked silicone)

The hydrolysis of alkyl trichlorosilane ($RSiCl_3$) gives a very complex cross-linking polymer.

2018, 2016, 2020, TU

(x) How are $[NPCl_2]_3$ and $[NPCl_2]_4$ produced? Discuss the salient feature of structure & bonding in $[NPCl_2]_3$?

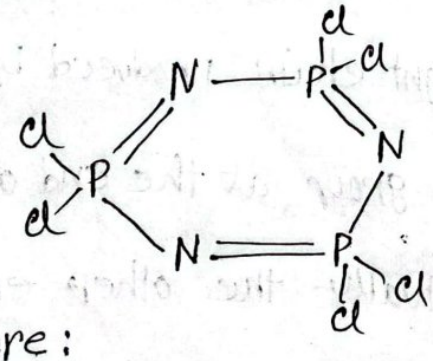
⇒ preparation of $[NPCl_2]_3$: (i) $3PCl_5 + 3NH_3 \rightarrow [NPCl_2]_3 + 9HCl$

(ii) $3PCl_5 + 3NH_4Cl \xrightarrow{145\text{-}160^\circ C} [NPCl_2]_3 + 12HCl$

preparation of $[NPCl_2]_4$: (i) $4PCl_5 + 4NH_3 \rightarrow [NPCl_2]_4 + 12HCl$

(ii) $4PCl_5 + 4NH_4Cl \xrightarrow{145\text{-}160^\circ C} [NPCl_2]_4 + 16HCl$

⇒ structure of $[NPCl_2]_3$:



feature:

the compound made up of tricovalent nitrogen & penta

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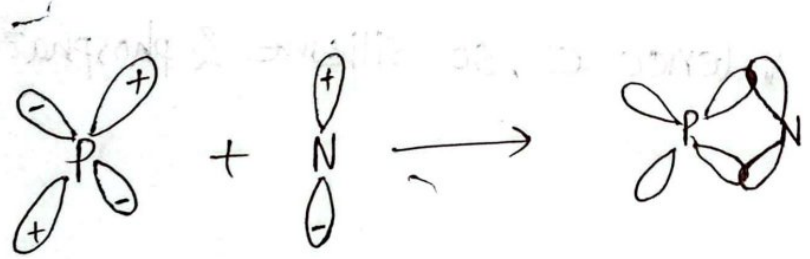
covalent ~~nitrogen~~ phosphorus atom. $[NPCl_2]_3$ have P & N atom with two Cl atoms which are attached to P atom. It consist of six-membered ring with altering P & N atoms. The ring is planar, with all atoms lying in the same plane.

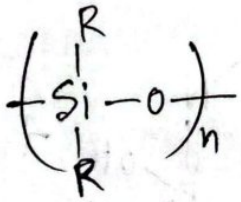
Bonding feature: In $[NPCl_2]_3$, the P-N & P=N bond lengths are equivalent as here delocalization occurs. Their bond lengths are ..1.58Å which is much shorter than the usual single bond length 1.77Å. Nitrogen is sp^2 hybridized & its filled p-orbital can overlap with P-atoms vacant d-orbitals that strengthens P-N bond & contributes to the delocalization of e^-.



2018

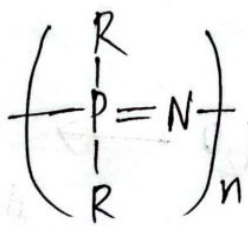
⊗ Silicone & phosphazene are isoelectronic - explain?

⇒ general structures are:



$(R_2SiO)_n$

(Silicone)



$(R_2PN)_n$

(phosphazene)

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Silicone & phosphazene are isoelectronic because their repeating units in the polymer chains have the same number of valence e^-, despite involving different elements.

The repeating unit in silicone is $-Si-O-$. its "Si" has 4 & "O" has 6 valence e^-. So total valence e^- per repeating unit is $= 4+6 = 10$.

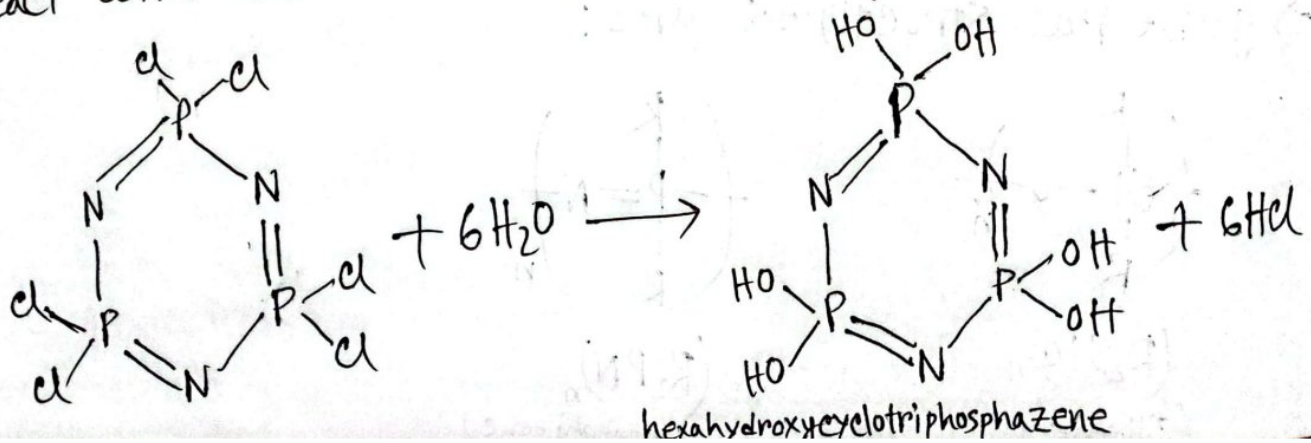
The repeating unit in phosphazene is $-P=N-$, its "P" has 5 & "N" has 5 valence e^-. So total valence e^- per repeating unit is $= 5+5 = 10$

As both $-Si-O-$ & $-P=N-$ repeating units have the same total number of valence e^-, so silicone & phosphazene are isoelectronic.

2018, 2017, TU

(*) what happens when $[NPCl_2]_3$ (i) is hydrolyzed with water, (ii) react with excess of NH_3 (iii) reacts with NaF & (iv) with $NaOCH_3$?

⇒ (i)



hexahydroxycyclotriphosphazene

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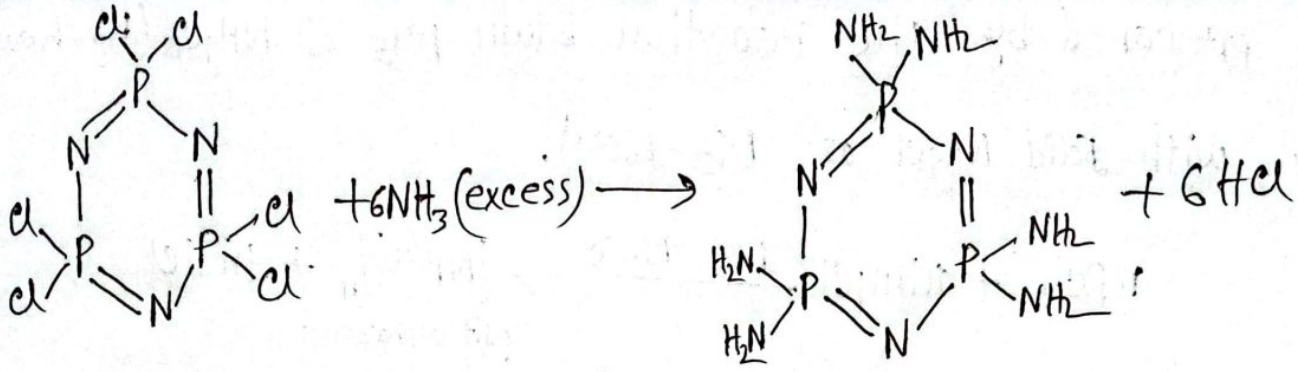
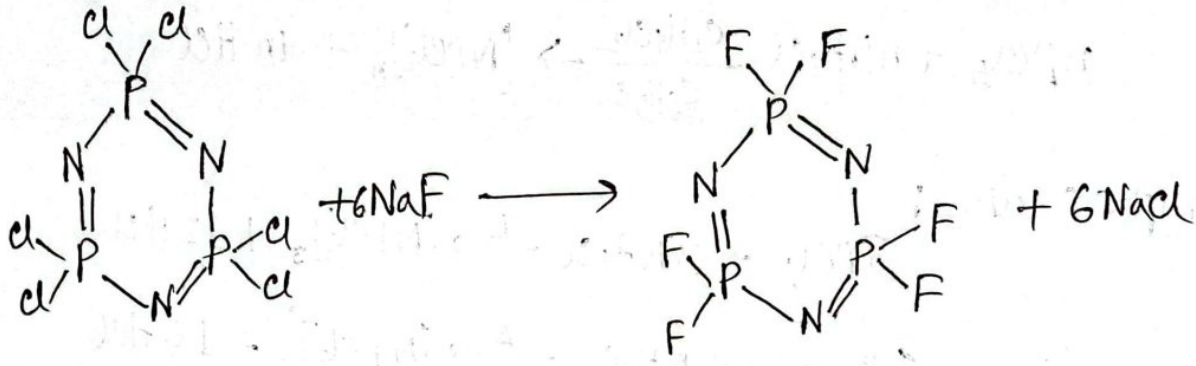
(II)



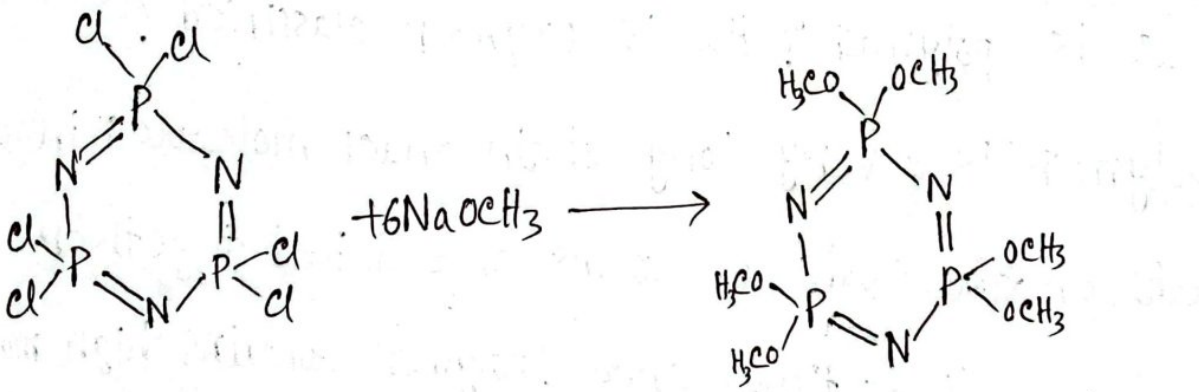
hexaminocyclotriphosphazene

(III)



hexafluorocyclotriphosphazene

(IV)



hexamethoxycyclotriphosphazene

2017, TU, 2022

(★) How phosphazene polymer obtained ?

⇒ phosphazene are a class of chemical compounds in which a "P" atom is covalently linked to a 'N' atom by a double bond & to three other atoms or radicals by single bonds. phosphazene can

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be prepared by the reaction btwn PCl_5 & NH_4Cl (by heating PCl_5 with solid NH_4Cl at 145-160°C).

$$n PCl_5 + n NH_4Cl \xrightarrow{145\text{-}160^\circ C} \underset{\text{polyphosphazene}}{[NPCl_2]_n} + 4n\ HCl$$

This reaction can be happend in presence of $C_2H_4Cl_2$ or C_6H_5Cl.

$$n PCl_5 + n NH_4Cl \xrightarrow[C_6H_5Cl]{C_2H_4Cl_2} [NPCl_2]_n + 4n\ HCl$$

examples are:

$$3 PCl_5 + 3 NH_4Cl \xrightarrow{\Delta} [NPCl_2]_3 + 12\ HCl$$

$$4 PCl_5 + 4 NH_4Cl \xrightarrow{\Delta} [NPCl_2]_4 + 16\ HCl$$

(★) 2016, 2020 What is polymer? How is polymer classified?

⇒ A polymer is a very long chain macromolecules in which hundreds or thousands of atoms are linked together to form a one dimentional array. polymer contains high molecular weight, repeating unit or monomer.

⇒ classification:

(A) Based on source: (I) Natural
(II) synthetic
(III) Semi- synthetic polymer.

(B) Based on structure: (I) linear, (II) Branched (III) cross-linked polymer.

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(c) Based on polymerization : ① Addition ② condensation polymers.

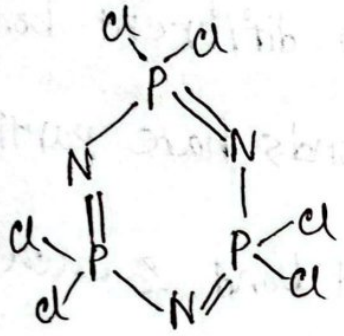
(d) Based on physical properties : ① thermoplastics ② thermosetting ③ Elastomers.

(e) Based on Monomer types : ① Homopolymers ② Co polymers

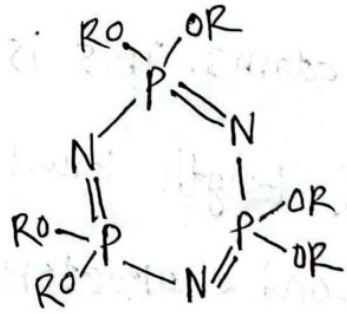
2015, 2022, 2021

⊛ What happens when $[NPCl_2]_3$ is heated to 250°C & react with ① RONa & ② RNH_2 ?

⇒

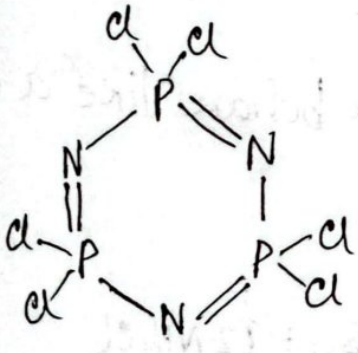
① 

$+6RONa \xrightarrow{250°C}$

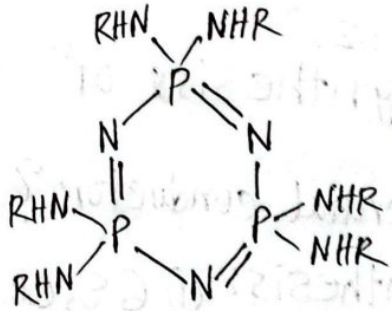


$+6NaCl$

hexaalkoxycyclotriphosphazene

② 

$+6RNH_2 \xrightarrow{250°C}$



$+6HCl$

hexachlorocyclotriphosphazene

Hexa-n-alkylaminocyclotriphosphazene

2017, TU

⊛ Explain the structure of S_4N_4 compound ?

⇒ synthesis reaction of tetrasulfur tetranitride (S_4N_4) is :

$$6S_2Cl_2 + 16NH_3 \longrightarrow S_4N_4 + S_8 + 12NH_4Cl$$

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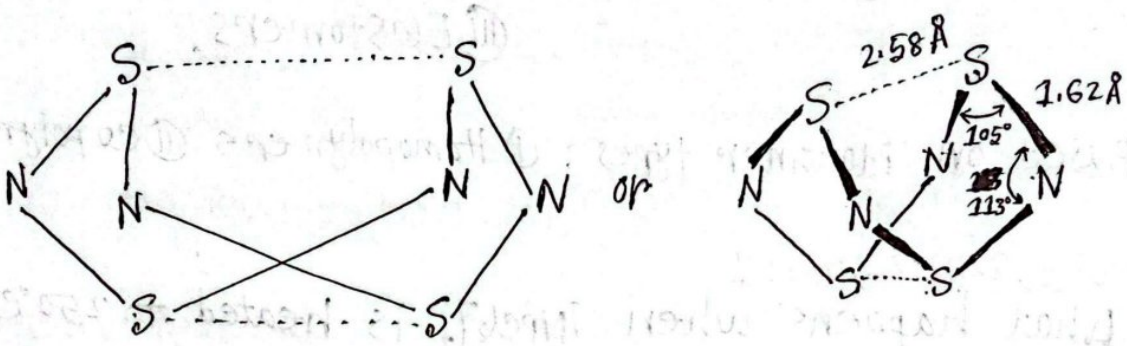
S_4N_4 is a cradle shape compound whose structure is different from S_8 ring which is crown shaped. structure:



S_4N_4 is Eight-membered ring structure with altering S & N atoms. This is cradle shape due to different bond angle & length btwn S & N atoms. S-N bonds have partial double bond character. S & N form covalent bond & each 'N' atom is bonded to two 'S' atom. It is not planar shaped molecule.

2022, TU
Ⓧ Synthesis of S_4N_4? How does $(SN)_x$ behave like a one dimential conductor?

⇒ synthesis: ① $6S_2Cl_2 + 16NH_3 \longrightarrow S_4N_4 + S_8 + 12NH_4Cl$

② $6SCl_2 + 16NH_3 \longrightarrow S_4N_4 + 2S + 12NH_4Cl$

⇒ the polymer $(SN)_x$ behaves like a one dimentional conductor due to its unique structure & electronic properties. $(SN)_x$ is composed of altering S & N atoms, forming a linear polymeric chain. The S-N bonds have partial double bond

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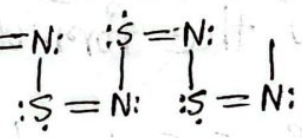
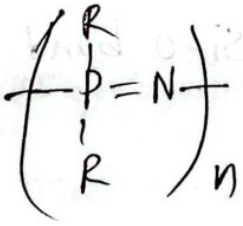
character & the π-e⁻ are delocalized along the chain. this delocalization allows e⁻ to move freely along the length of the chain, facilitating conductivity in one dimension. e⁻ mobility is high along the polymer chain but negligible perpendicular to the chain. the directional conduction makes $(SN)_x$ effectively a one-dimentional conductor. Due to the delocalized π-e⁻ system, $(SN)_x$ exhibits metallic conductivity at low temperatures. $(SN)_x$ lewis structure:



2022

Discuss the bonding of phosphazene?

⇒ Phosphazene backbone is made up of alternating single & double ~~bonds~~ P-N bonds.



$(R_2PN)_n$

The P-N bond in phosphazenes has both σ & π character. The π bond is formed by the overlap of sp^3 orbitals of 'P' & sp^2 orbitals from 'N'. The π bond is formed by the

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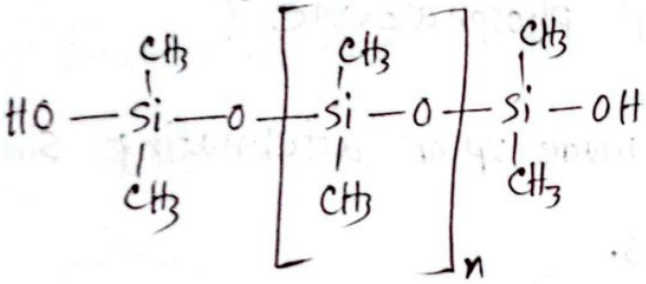
overlap of unhybridized p orbitals of 'N' with ~~vacant~~ vacant d orbitals of 'P' allowing for delocalization. phosphazene exhibits resonance, where the e^- in the π-system are delocalized across the P=N bonds. The presence of π bond results in a shorter bond length btwn 'P' & 'N' compared to a single P-N bond.

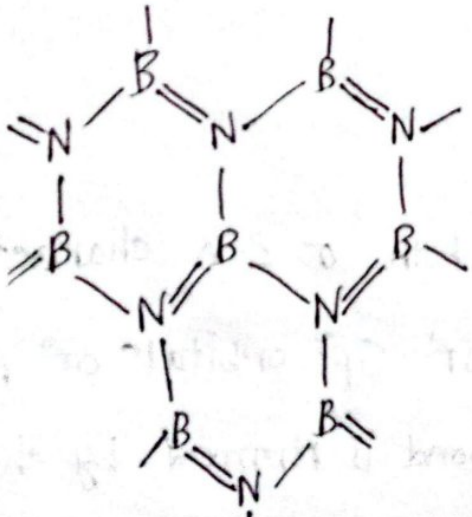
2016, 2020, 2021

Ⓐ Draw the formulae of one polymer having Si-O, B-N & S-S bond?

⇒



Silicone polymer having Si-O bond



Boron nitride, $(BN)_x$ polymer having B-N bond

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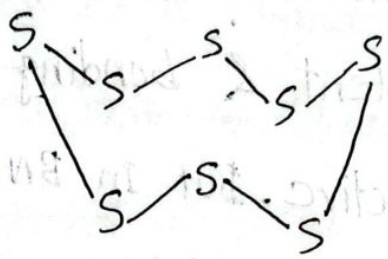
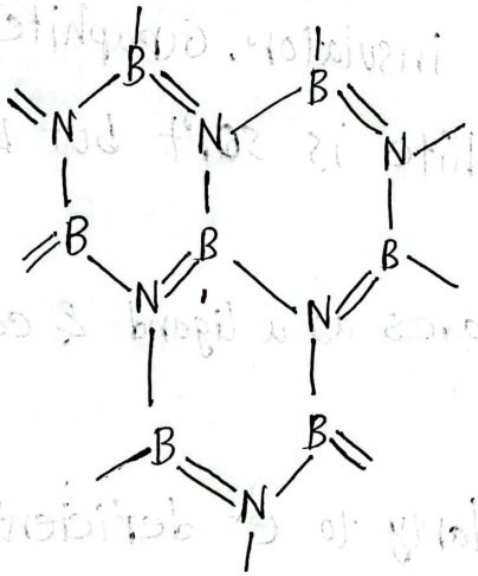


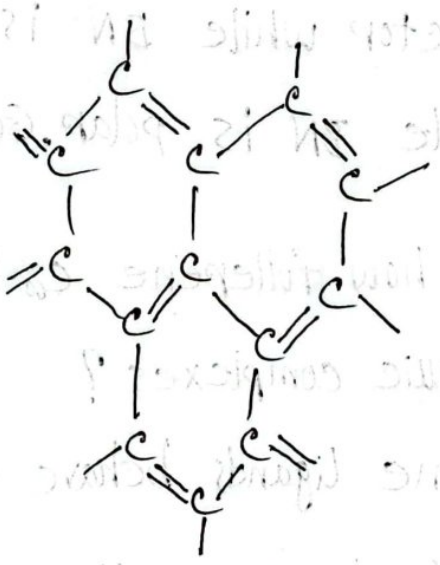
S_8 polymer having S-S bond

(★) 2021 How does $(BN)_x$ resemble with & differ from graphite?

⇒



Boron nitride, $(BN)_x$



Graphite

$(BN)_x$ & graphite have similar structure both have planes of hexagons. Both materials possess a layered structure, where individual layers are held together by weak van der walls forces.

Despite ~~similarities~~ their structural similarities, BN & graphite differ signinificantly: BN consist of alternating

B & N atom but graphite consists of only carbon atom. In graphite all 'C' atoms are equivalent & bonding involves delocalized π-e^- making it conductive. But In BN, the BN bonds are partially ionic due to the difference in electronegativity btwn B & N. This results in less π-e^- delocalization & poor electrical conductivity. So graphite is good conductor while BN is an insulator. Graphite is non-polar while BN is polar. Graphite is soft but BN is hard

2021
⊛ Explain how fullerene C_{60} behaves as a ligand & cage organometallic complexes?

⇒ Fullerene ligands behave similarly to e^- deficient alkenes and coordinate in a dihapto fashion. This building occurs on the junction points of two 6-membered rings. Fullerene C_{60} is composed of 60 carbon atoms arranged in a spherical shape with alternating pentagonal & hexagonal rings. Depending on the metal & its oxidation state, C_{60} can act as a ~~acceptor~~ π-acceptor ligand & σ donor ligand. The delocalized π-e^- system of C_{60} can accept e^- density from metal.

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centers, acting as a π-acid ligand. When metal is e⁻-deficient, then C_{60} can also donate e⁻ density to the metal center through σ-bonding interactions.

C_{60} can form a variety of organometallic complexes with different metal centers. such as;

η^2-C_{60} complexes: In these complexes, the metal center is coordinated to two adjacent 'C' atoms of the C_{60} case through a π-bond.

η^6-C_{60} complexes: In these complexes, the metal center is coordinated to two adjacent carbon atoms of the C_{60} case, forming a π-complex.

Endohedral complexes: In these complexes, a metal atom or cluster is encapsulated within the C_{60} cage.

$$[(C_6H_5)_3P]_2Pt(\eta^2\text{-}C_2H_4) + C_{60} \longrightarrow [(C_6H_5)_3P]_2Pt(\eta^2\text{-}C_{60})$$

(*) 2017, 2020, 2016
Write chemistry of fullerenes? Or short note about Fullerene, C_{60}?

⇒ Fullerene is any molecule composed entirely of carbon, in the form of a hollow sphere, ellipsoid or tube. spherical

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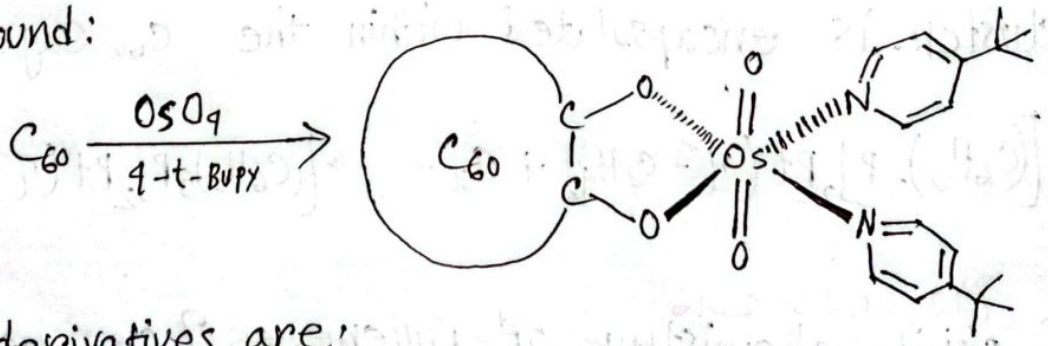
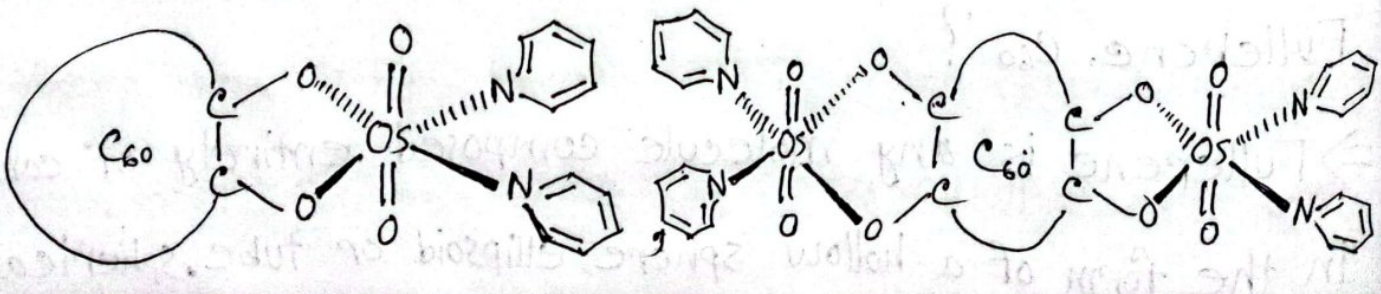
fullerenes are also called buckyballs and cylindrical ones are called carbon nanotubes or buckytubes. physical properties are: Insoluble in H_2O, cannot conduct electricity, soft & slippery, Brittle, low melting point. Fullerenes carbon is sp^2 hybridized. C_{60} is composed of 12 pentagons and 20 hexagons. It contains 60 single bond & 30 double bonds. Each carbon is part of one pentagon & two hexagons, each has two single bonds & one double bond for the traditional carbon valence of four. The pentagonal rings contain only single bonds; double bonds have a shorter bond length & lead to instability in the pentagonal ring. The first derivative structure of C_{60} was the remarkable osmium compound:



some other derivatives are:



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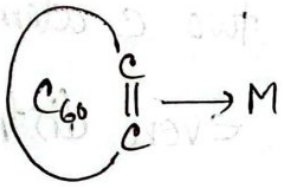
C_{60} can behave as an alkene towards transition metals, a side on η_2 connection to either pt or Ir is made by π-e of a C=C group.

$$C_{60} + (\eta_2\text{-}C_2H_4)Pt(PPh_3)_2 \longrightarrow C_2H_4 + (\eta_2\text{-}C_{60})Pt(PPh_3)_2$$

$$C_{60} + Ir(CO)Cl(PPh_3)_2 \longrightarrow (\eta_2\text{-}C_{60})Ir(CO)Cl(PPh_3)_2$$

These compounds can be formulated as:



Fullerene aggregates are bonded by van der waals forces so foreign elements are readily intercalated in the lattice.

C_{60}'s application: In Antioxidant & antiflammatory medicine, In water purification, to form superconductors, used in cosmetics, In drug delivery, for energy storage in Battery.

Chemistry: Fullerenes are stable but not totally unreactive every atom sp^2 hybridized. other atom can be trapped inside fullerenes to form inclusion compounds known as endohedral fullerenes.

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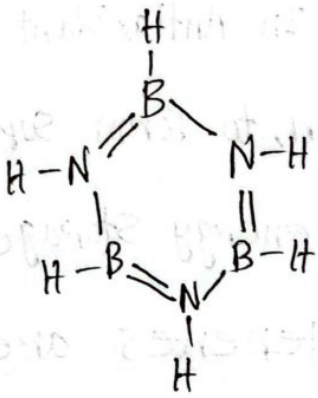
2015
⊛ B-N bond in borazine is slightly polar but c-c bond in benzene is non polar - explain?

⇒ The electronegativity difference btwn B & N creats a partial +ve ~~-ve~~ charge on 'B' & partial -ve charge on 'N'. This partial +ve & -ve charge makes the B-N bond slightly polar in borazine.

But in benzene, C-C bond's two C atom have same electronegativity. This results even distribution of e⁻ density within the bond making it non-polar.

2022
⊛ Draw the structure of Borazine & explain its non-aromatic character?

⇒ Structure of Borazine:



⇒ Borazine is often called as inorganic benzene due to its structural similarity to benzene. The B-N bond's ~~have~~ ~~significantly~~ significant polarity due to the difference in electronegativity btwn B & N. This disrupts

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the delocalization of e^- within the ring, hindering aromaticity. The e^- density is not evenly distributed across the ring. 'N' atom tend to pull e^- density toward themselves, further disrupting delocalization. The π-e are localized on the 'N' atom, preventing them from delocalizing across the entire ring.

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Lanthanoids & Actinoids

RAJIB
Che-50

⊛ 2021 off all the Lanthanides Yb (z=70) shows the highest ionization energy for the formation of +3 ion – why?

⇒ due to its electronic configuration & stability of its partially filled subshells.

When Yb loses two e⁻, it achieves a stable, ~~half~~ filled 4f subshell, which is stable configuration. Removing 3rd e⁻ from Yb^{2+} disrupts this stable configuration & requires more energy. This is because the remaining 3rd e⁻ is tightly held within ~~half~~ filled 4f subshell, experiencing a strong effective nuclear charge.

$$Yb = [Xe]\,4f^{14}\,6s^{2} \;;\; Yb^{2+} = [Xe]\,4f^{14} \;;\; Yb^{3+} = [Xe]\,4f^{13}$$

$4f^{14}$ configuration is exceptionally stable, making it more difficult to remove an additional e⁻ to form Yb^{3+}.

⊛ 2024, 2018, 2022 off all $Ln(OH)_3$, basic character of $Ce(OH)_3$ is maximum but that of $Lu(OH)_3$ is minimum?

⇒ Due to lanthanide contraction. Ce is one of the early lanthanides, so it has a larger ionic radius & lower e⁻ density.

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The larger ionic radius allows the $\bar{O}H$ to be less tightly held making $Ce(OH)_3$ more basic. But Lu is the last lanthanide so it has smallest ionic radious & highest charge density. This smaller radius tightly hold the OH^- & makes $Lu(OH)_3$ less basic.

2020, 2016, 2017

⊛ what is lanthanide contraction? what is the consequence of lanthanide contraction?

⇒ As the atomic number increases each succeeding element contains one more e^- in the 4f orbital & one proton in the nucleus. The 4f e^- as ineffective in screening the outer e^- from the nucleus causing imperfect shielding. As a result there is a gradual increase in the nucleus attraction for the outer e^- consequently gradual decrease in size occur. This is called Lanthanide contraction.

or,

Lanthanide contraction refers to the decrease in atomic size of the element in which e^- fill the f-subshell. Since the f subshell is not shielded, the atomic size will decrease as the nuclear charge still increase.

⇒ consequences: ① Basicity of ions decreases from La^{3+} to Lu^{3+}

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so their oxides & hydroxides ar less basic.

(ii) As the atomic size decreases across the series the elements following lanthanides show change in their physical properties.

(a) the ionization potential increases. (b) Metallic nature decreases (c) electropositivity decreases (d) Electronegativity increases (e) Tendency to form complexes increases.

(iii) Increased covalent character as size decreases (cation).

(iv) Lanthanide contraction makes the properties of lanthanides very similar, which complicates their seperation & purification.

(x) 2017, 2016, 2020 Name the various methods for extraction of lanthanides & discuss any one method for their extraction ?

⇒ various methods are: (i) Repeated fractional crystallization method (ii) complex formation method (iii) fractional decomposition of oxy salt method (iv) solvent extraction method (v) change in oxidation state method (vi) fractional precipitation method (vii) Ion exchange method.

⇒ Ion exchange method: An aqueous sln of the mixture of

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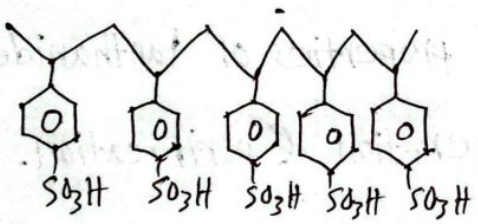
Lanthanide ions (Ln^{3+}_{aq}) is introduced into a column containing a synthetic cation exchange resin such as DOWAX-50. The resin is the sulphonated polystyrene containing $-SO_3H$ as the functional group. As the sln of the mixture moves through the column, Ln^{3+}_{aq} ions replace H^+ ions of the resin & get themselves fixed on it & seperated.

$$Ln^{3+}_{(aq)} + 3H^+(resin) \longrightarrow Ln(resin)_3 + 3H^+_{(aq)}$$



sulphonated polystyrene

2017
(*) What are lanthanides? write down their common electronic configuration?

⇒ The lanthanides (15 elements) are the elements produced as the 4f(1-14) sub level is filled with electrons.

Lanthanides are a group of 15 chemical elements with atomic number 57 to 71, following elements are called lanthanum. The series is called lanthanide series.

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Lanthanided electronic configuration → $[Xe]\ 4f^{1-14}\ 5d^{0-1}\ 6s^{2}$

Actinides electronic configuration → $[Rn]\ 5f^{1-14}\ 6d^{0-1}\ 7s^{2}$

2016, 2015

(**) State the color of actinide ions?

The colors of actinide ions arise due to electronic transition, particularly within the 5f orbitals, or btwn the 5f & other orbitals (e.g. 6d or 6p). These transitions are influenced by the e⁻ configuration, oxidation state & the surrounding chemical environments (ligands).

5f-5f transitions: The most common type of electronic transitions in actinides, responsible for their characteristic color.

Number of 5f e⁻: Influences the color intensity. Ions with zero ($5f^{0}$) or seven ($5f^{7}$) 5f electrons are typically colorless.

oxidation states: Actinides exhibite a range of oxidation states, commonly +3, +4, +5 & +6. Higher oxidation states often result in more intense & varied colors.

Ligand fields effects: The nature of the ligands surrounding the

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actinide ion can also affect the energy levels of the 5f orbitals & thus influence the color. Ex. U^{3+} → Red, Np^{3+} → purple, Am^{3+} → pink, Cm^{3+} → colorless, U^{4+} → Green, Np^{4+} → yellow green.

2015, 2018

⊛ Actinide show higher oxidation state than lanthanide-why?

⇒ Lanthanides → $[Xe]\ 4f^{1-14}\ 5d^{0-1}\ 6s^{2}$ → +3 (dominant), +2, +4

Actinides → $[Xe]\ 5f^{1-14}\ 6d^{0-1}\ 7s^{2}$ → +3, +4, +5, +6, (+7 in some)

(i) In actinides energy gap btwn 5f, 6d & 7s orbitals is smaller which allow multiple e^- to be removed resulting higher ox. state.

(ii) The high atomic numbers of actinides lead to significant relativistic effects, which further, reduce energy gap btwn 5f, 6d, 7s.

(iii) 5f orbitals are less effectively shielded by outer e^- compared to the 4f orbitals. This weaker shielding allows the nuclear charge to penetrate more effectively, Influencing the energy levels of the 5f e^- & contributing to the wider range of oxidation states. so actinide show higher o. state.

2015, 2018

⊛ Discuss the methods for the seperation of lanthanides from Monazite?

⇒ Monazite is a reddish-brown phosphate mineral containing rare earth metals. Monazite's general formula $(La, Ce, Nd, Sm)PO_4$.

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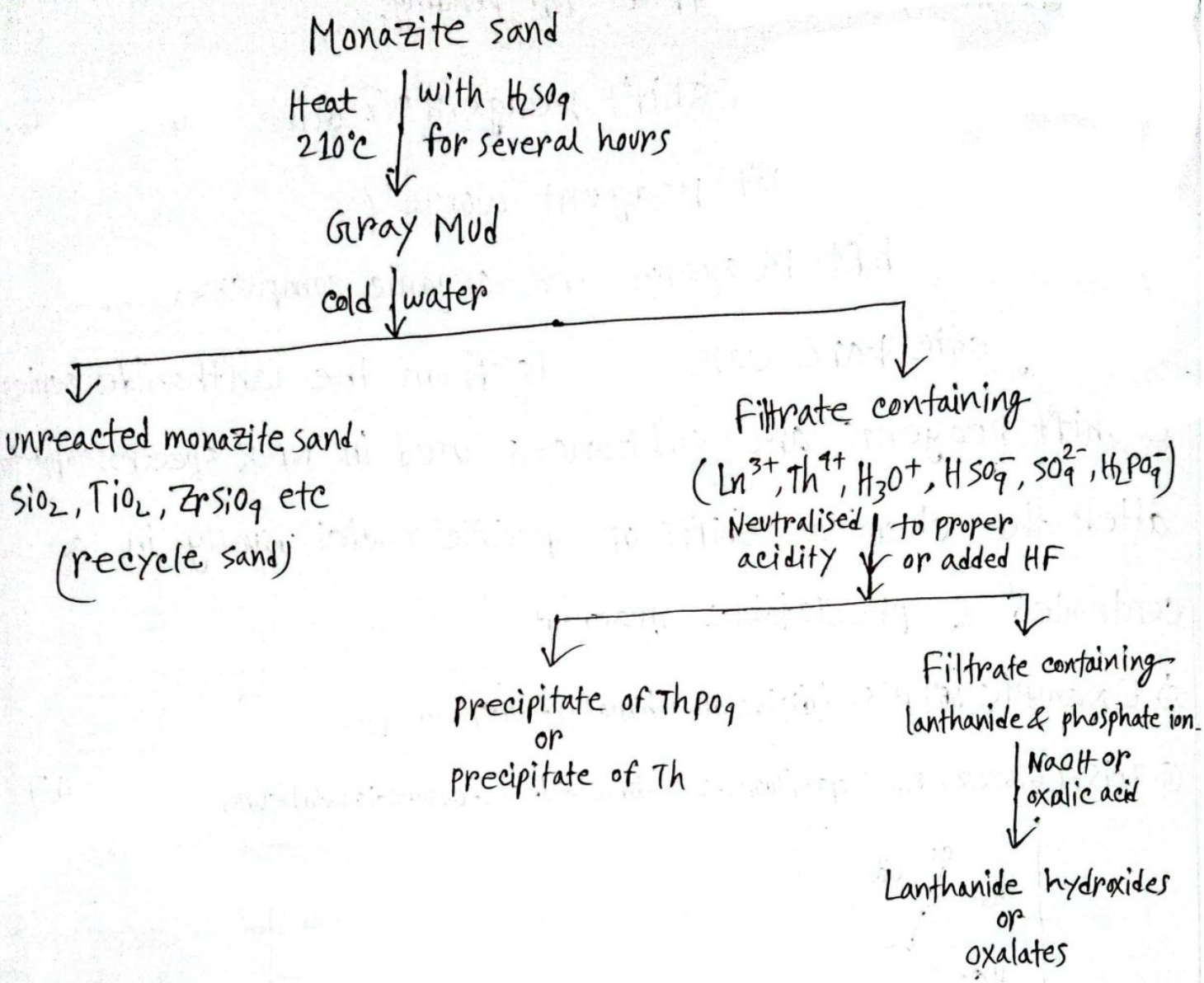

2015, 2018

Q Why 4f e⁻ of lanthanide do not take part in chemical bonding?

⇒ In lanthanides, the 4f orbitals are well shielded by the larger 5d & 6s orbitals & are deep inside the atoms. so 'f' orbitals do not participate in any bonding. This shielding reduces the interaction btwn the 4f e⁻ & the valence e⁻ of other atoms. The 4f orbitals experience lanthanide contraction, leading to a decrease in atomic size. this contraction reduces

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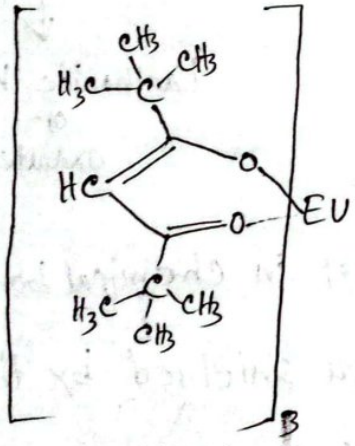
the accessibility of the 4f e⁻ for bonding.

⊛ 2022 What are chemical shift reagents? Give two examples. How chemical shift reagent work?

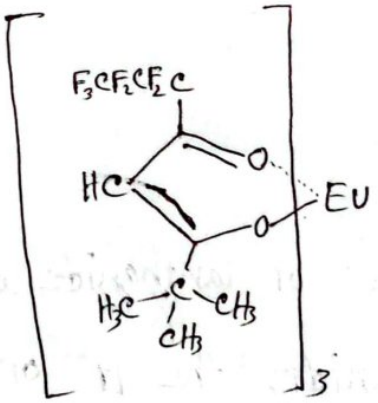
⇒ chemical shift reagents are organic complexes of paramagnetic rare earth metals from the lanthanide series. C. shift reagents are substances used in NMR spectroscopy to alter the chemical shifts of specific nuclei, usually in a controlled & predictable manner.

⇒ Example: ① Tris-(dipivalomethanato) europium [Eu(dpm)$_3$]

② Tris-(6,6,7,7,8,8,8-heptafluoro-2,3-dimethyl-3,5-octanedionato) europium [Eu(fod)$_3$]



Eu(dpm)$_3$



Eu(fod)$_3$

⇒ The chelates forms a complex with the functional group of the substrate & induce spectral simplification in the NMR spectrum. these chelates are lewis acid & the lone pair of e⁻ of the functional group coordinate with chelate (Eu^{3+})

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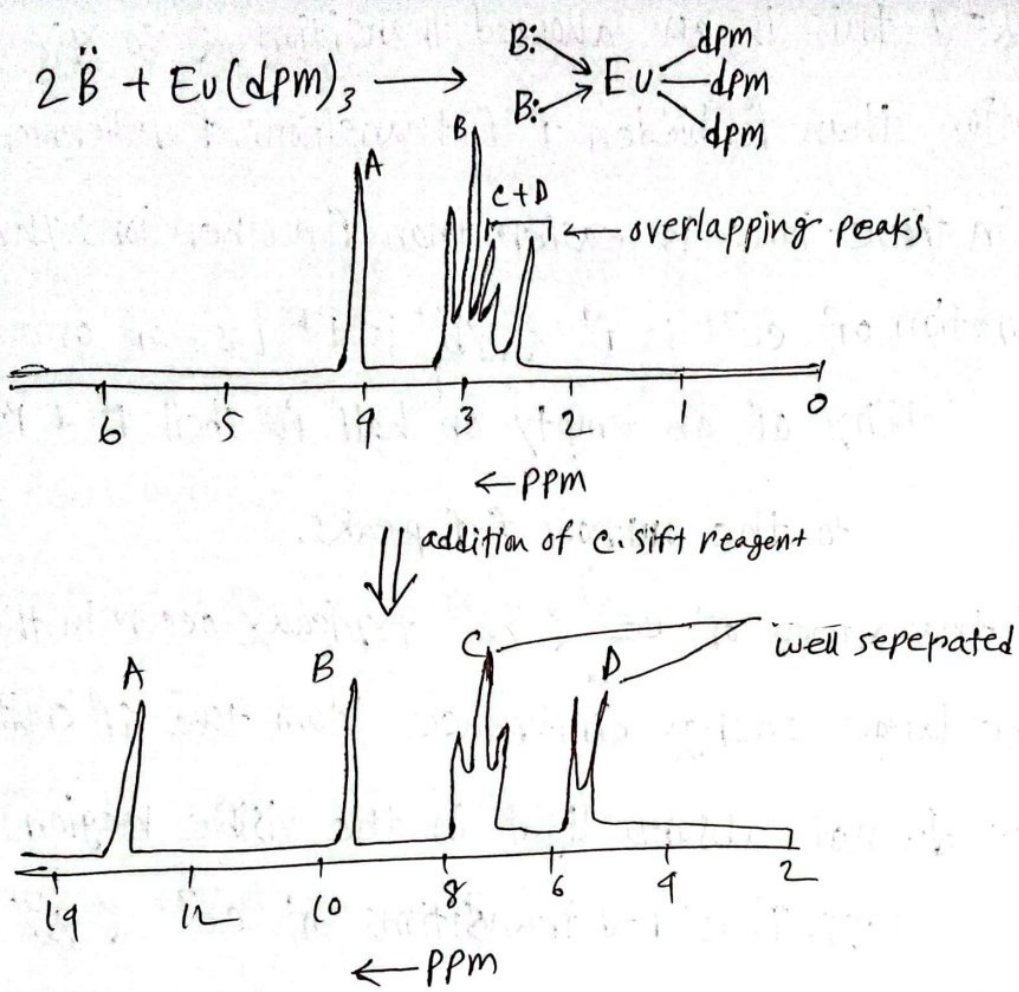

c.s. reagents are paramagnetic compounds that are used in NMR spectroscopy to simplify & interpret NMR spectra. c.s.r. induce large changes in the chemical shift of molecule by attaching a paramagnetic ion to a ~~fix~~ ligand that complexes with the molecule.

2022
(☆) Ce^{3+} & Yb^{3+} are colorless but show strong absorption in UV-region - explain?

⇒ Ce^{3+} & Yb^{3+} are colorless because they do not absorb in the visible region. But they show strong absorption in UV because of transition from 4f to 5d. absorption is very strong for two

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reasons since $\Delta l = 1$ this is an allowed transition & so gives stronger absorption than forbidden f-f transitions. Furthermore promotion of e^- in these ions is easier than for other ions. The electronic configuration of ce^{3+} is f^1 & Yb^{3+} is f^{13}. Loss of one e^- gives the extra stability of an empty or half full shell f-d peaks are broad, in contrast to the narrow f-f peaks.

~~The strong~~ f-f transitions of ce^{3+} & Yb^{3+} typically occur in the UV region due to the large energy difference btwn the 4f orbitals. Since transitions do not absorb light in the visible region, so the compounds colorless. The f-d transitions of ce^{3+} & Yb^{3+} are responsible for strong absorption in the UV region. 4f-5d transitions require more energy so occur in UV region & show strong absorption in UV region.

2022 Complete the reaction! $Lu(OH)_3 + NaOH \longrightarrow$?

$\Rightarrow Lu(OH)_3 + 3NaOH \longrightarrow Na^+ + [Lu(OH)_6]^{3-}$

2022 All but Lu^{3+} of the lanthanide ions show absorption in the visible or near UV spectrum & their colors are pale - explain?

$\Rightarrow Lu^{3+}$ (Lutetium ion) has completely filled 4f orbitals ($4f^{14}$). As a result, there is no available 4f-4f transitions, so it

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does not absorb light in the visible or near-uv region & appears colorless. There is no available energy levels for electronic transition when absorbing light in the visible or uv-range as 4f is filled. So no light in this region absorbed & Lu^{3+} compounds will be colorless.

2021
⁂ covalent radii of Mo ($Z=42$) & W ($Z=74$) are approximately the same - why? [1.29 Å & 1.30 Å respectively].

⇒ Due to lanthanide contraction. Tungsten comes after the lanthanides ($Z=57$ to $Z=71$), which have poorly shielded 4f e^-. The 4f orbitals in these elements do not shield the nuclear charge effectively, leading to a stronger effective nuclear charge experienced by the outer e^-. This stronger e.n. charge causes the outer orbitals to contract, resulting in a smaller atomic size than expected.

Mo is in 4d series & W is in 5d series. So Normally moving down in group leads to increas size but lanthanide contraction reduces this expected increase for W. Both are in group 6.

2021
⁂ Mention the fluorides of uranium & gives reactions for their formation from UO_2?

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⇒ Fluorides of uranium : UF_4 , UF_5 , UF_6 .

Formation of UF_4 :

① Reaction with HF acid : $UO_2(s) + 4HF \longrightarrow UF_4 + 2H_2O$

② Reaction with Elemental Fluorine (F_2) : $UO_2(s) + 2F_2 + H_2 \longrightarrow UF_4 + H_2O$

Formation of UF_5 :

① Reaction with Fluorine gas : $UO_2 + 3F_2 \longrightarrow UF_6 + O_2$

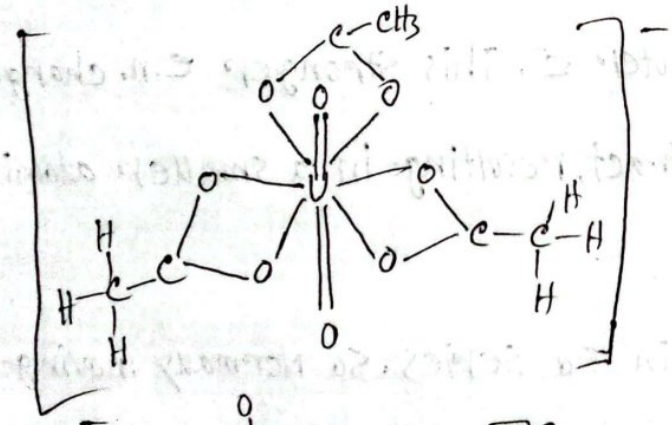
Formation of UF_5 :

① $UO_2 + 4HF \longrightarrow UF_4 + 2H_2O$

$2UF_4 + F_2 \longrightarrow 2UF_5(s)$

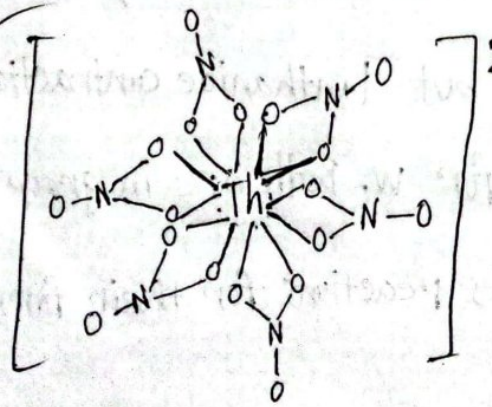
⊗ 2021 Draw structure & mention co-ordination number of the central metals : ① $[UO_2(CH_3COO)_3]^-$ & ② $[Th(NO_3)_6]^{2-}$?

⇒ ①



→ co-ordination number = 6

②



→ co-ordination number = 12

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(2020) How do the lanthanide complexes act as NMR shift reagent?

⇒ because of their unique electronic properties, which induce changes in the chemical shifts of nearby nuclei in an NMR spectrum. Many lanthanide ions are paramagnetic due to unpaired e^- in their 4f orbitals. These unpaired e^- generate a strong magnetic field that ~~interacts~~ causes an induced shift in the resonant frequency of nearby nuclei, either upfield (shielding) or downfield (deshielding). When lanthanide complex is added to the sample, it binds to specific sites. above induced shift depends on the distance btwn the lanthanide ion & the nuclei. The lanthanide ion's field creates an anisotropic environment, meaning the chemical shift changes are dependent on the relative positions of atoms in the molecule.

(2018) Why are Yb^{2+} & Eu^{2+} more stable than Yb^{3+} & Eu^{3+} ions?

⇒ $Eu^{2+} \longrightarrow [Xe]\ 4f^7$

$Yb^{2+} \longrightarrow [Xe]\ 4f^{14}$

Eu^{2+} have half filled & Yb^{2+} have full-filled f orbital. These configurations are considered to be more stable due to exchange energy & pairing energy considerations. so Yb^{2+} & Eu^{2+} more stable.

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2018

⊗ Why La^{3+}, Ce^{4+} & Lu^{3+} are diamagnetic but all other Lanthanides ion are paramagnetic?

⇒ $La^{3+} \longrightarrow [Xe]\ 4f^0$

$Ce^{4+} \longrightarrow [Xe]\ 4f^0$

$Lu^{3+} \longrightarrow [Xe]\ 4f^{14}$

In these three ions there is no unpaired e^- so they are diamagnetic. But all other Lanthanides have partially filled f orbital, leading to unpaired e^-. These unpaired e^- contribute in paramagnetism. Ex: $Pr^{3+} \rightarrow [Xe]\ 4f^2$ (2 unpaired)

$Nd^{3+} \rightarrow [Xe]\ 4f^3$ (3 unpaired).

2017

⊗ Ln^{3+} ions forms stable complexes with oxygen rich chelating Ligands? - Explain?

⇒ Ln^{3+} ions are classified as hard acids because of their small size & high charge density. oxygen rich chelating ligands are hard bases due to high electronegativity & small size of oxygen atoms. Hard acids interact strongly with hard bases, leading to stable complexes.

oxygen rich ligands provide negative charge density, which interacts effectively with the positively charged Ln^{3+}.

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chelating ligands have multiple donor atoms that can simultaneously bind to the Ln^{3+} ion. This multiple coordination enhance stability due to stable ring structure. High charge (+3) of Ln^{3+} ions creates a strong electrostatic attraction with negatively charged oxygen donor atoms, enhancing complex stability

2017

(★) Ln^{3+} do not form complexes readily like transition metals - why?

⇒ Lanthanides have a stable electronic configuration with their 4f orbitals deeply buried beneath the 5s, 5p & 6s orbitals. These 4f orbitals are not spatially extended & do not participate significantly in bonding. In contrast, transition metals have partially filled 3d, 4d or 5d orbitals, which are more available for covalent bonding with ligands.

Ln^{3+} ions tend to have high co-ordination numbers (8-12) due to their large ionic radii, which limits their ability to form smaller, well-defined complexes. Transition metal typically have lower co-ordination numbers (4-6), allowing them to form complexes. Lanthanides form complexes more slowly compared to transition metals because their reactions are dominated by ionic interactions, which are less directional & less specific

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Lanthanide ions have relatively large ionic radii compared to transition metal ions. This larger size can hinder the approach of ligands & do not form complexes readily.

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